

Theory of Mathematics

Volume V: Commutative Algebra and Homological Methods

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Part I

Rings and Ideals

Chapter 1

Rings, Ideals and Quotients

Commutative algebra begins with rings, ideals, and quotients. Ideals are exactly the additive subgroups that can serve as kernels of ring homomorphisms, and quotient rings convert congruence conditions into algebraic objects.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

1.1 Commutative rings

A commutative ring has associative addition and multiplication, a multiplicative identity, and commutative multiplication.

1.2 Ideals

An ideal is an additive subgroup stable under multiplication by arbitrary ring elements.

1.3 Generated ideals

The ideal generated by elements $a_1, \dots, a_n$ consists of finite combinations $r_1 a_1 + \dots + r_n a_n$.

1.4 Sums and intersections

Sums give the smallest ideal containing two ideals, while intersections give their greatest common subideal.

1.5 Quotient rings

For an ideal $I \subset A$, the quotient A/I has cosets as elements and multiplication induced from A .

1.6 Ring homomorphisms

Kernels are ideals and images are subrings; quotienting by the kernel isolates the injective part of a map.

1.7 Correspondence of ideals

Ideals of A/I correspond to ideals of A containing I .

1.8 Examples

Polynomial rings, integer residue rings, coordinate-style quotients, and product rings provide the main laboratory.

1.9 Core structural results

Theorem 1.1 (First isomorphism theorem). *For a ring homomorphism $\varphi : A \rightarrow B$, there is a natural isomorphism $A/\ker \varphi \cong \text{im } \varphi$.*

Proof. Send the class of a to $\varphi(a)$. Equality of images is exactly equality modulo the kernel, so the map is well-defined and injective; surjectivity onto the image is immediate. $\square$

Theorem 1.2 (Correspondence theorem). *Ideals of A/I are in inclusion-preserving bijection with ideals $J \subset A$ satisfying $I \subseteq J$.*

Proof. Take inverse image under the quotient map in one direction and quotient J/I in the other. The two constructions are inverse by direct inspection. $\square$

Theorem 1.3 (Third isomorphism theorem). *If $I \subseteq J \subseteq A$ are ideals, then $(A/I)/(J/I) \cong A/J$.*

Proof. Apply the first isomorphism theorem to the homomorphism $A/I \rightarrow A/J$ sending $a + I$ to $a + J$; its kernel is J/I . $\square$

1.10 Worked examples

Example 1.4 (Commutative rings diagnostic). Give a rigorous worked analysis of Commutative rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A commutative ring has associative addition and multiplication, a multiplicative identity, and commutative multiplication. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 1.5 (Ideals diagnostic). Give a rigorous worked analysis of Ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An ideal is an additive subgroup stable under multiplication by arbitrary ring elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 1.6 (Generated ideals diagnostic). Give a rigorous worked analysis of Generated ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The ideal generated by elements $a_1, \dots, a_n$ consists of finite combinations $r_1a_1 + \dots + r_na_n$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 1.7 (Sums and intersections diagnostic). Give a rigorous worked analysis of Sums and intersections. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Sums give the smallest ideal containing two ideals, while intersections give their greatest common subideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

1.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 1.8 (Commutative rings diagnostic). Give a rigorous worked analysis of Commutative rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A commutative ring has associative addition and multiplication, a multiplicative identity, and commutative multiplication. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 1.9 (Ideals diagnostic). Give a rigorous worked analysis of Ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An ideal is an additive subgroup stable under multiplication by arbitrary ring elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 1.10 (Generated ideals diagnostic). Give a rigorous worked analysis of Generated ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The ideal generated by elements $a_1, \dots, a_n$ consists of finite combinations $r_1a_1 + \dots + r_na_n$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 1.11 (Sums and intersections diagnostic). Give a rigorous worked analysis of Sums and intersections. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Sums give the smallest ideal containing two ideals, while intersections give their greatest common subideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 1.12 (Quotient rings diagnostic). Give a rigorous worked analysis of Quotient rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For an ideal $I \subset A$, the quotient A/I has cosets as elements and multiplication induced from A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 1.13 (Ring homomorphisms diagnostic). Give a rigorous worked analysis of Ring homomorphisms. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Kernels are ideals and images are subrings; quotienting by the kernel isolates the injective part of a map. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 1.14 (Correspondence of ideals diagnostic). Give a rigorous worked analysis of Correspondence of ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ideals of A/I correspond to ideals of A containing I . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 1.15 (Examples diagnostic). Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Polynomial rings, integer residue rings, coordinate-style quotients, and product rings provide the main laboratory. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 1.16 (First isomorphism theorem proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For a ring homomorphism $\varphi : A \rightarrow B$, there is a natural isomorphism $A/\ker \varphi \cong \operatorname{im} \varphi$.

Solution. Send the class of a to $\varphi(a)$. Equality of images is exactly equality modulo the kernel, so the map is well-defined and injective; surjectivity onto the image is immediate. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 1.17 (Correspondence theorem proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Ideals of A/I are in inclusion-preserving bijection with ideals $J \subset A$ satisfying $I \subseteq J$.

Solution. Take inverse image under the quotient map in one direction and quotient J/I in the other. The two constructions are inverse by direct inspection. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 1.18 (Third isomorphism theorem proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $I \subseteq J \subseteq A$ are ideals, then $(A/I)/(J/I) \cong A/J$.

Solution. Apply the first isomorphism theorem to the homomorphism $A/I \rightarrow A/J$ sending $a + I$ to $a + J$; its kernel is J/I . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 1.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Commutative algebra begins with rings, ideals, and quotients. Ideals are exactly the additive subgroups that can serve as kernels of ring homomorphisms, and quotient rings convert congruence conditions into algebraic objects. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

1.12 Exercises with complete solutions

Exercise 1.20. State and apply the central principle behind Commutative rings. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A commutative ring has associative addition and multiplication, a multiplicative identity, and commutative multiplication. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 1.21. State and apply the central principle behind Ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An ideal is an additive subgroup stable under multiplication by arbitrary ring elements. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 1.22. State and apply the central principle behind Generated ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The ideal generated by elements $a_1, \dots, a_n$ consists of finite combinations $r_1 a_1 + \dots + r_n a_n$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 1.23. State and apply the central principle behind Sums and intersections. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Sums give the smallest ideal containing two ideals, while intersections give their greatest common subideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 1.24. State and apply the central principle behind Quotient rings. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For an ideal $I \subset A$, the quotient A/I has cosets as elements and multiplication induced from A . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 1.25. State and apply the central principle behind Ring homomorphisms. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Kernels are ideals and images are subrings; quotienting by the kernel isolates the injective part of a map. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 1.26. State and apply the central principle behind Correspondence of ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Ideals of A/I correspond to ideals of A containing I . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 1.27. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Polynomial rings, integer residue rings, coordinate-style quotients, and product rings provide the main laboratory. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 2

Prime and Maximal Ideals

Prime and maximal ideals encode the irreducible quotient structures of a commutative ring. Prime ideals detect domains, maximal ideals detect fields, and maximal ideals are automatically prime.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

2.1 Prime ideals

A proper ideal $\mathfrak{p}$ is prime when $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$.

2.2 Maximal ideals

A proper ideal $\mathfrak{m}$ is maximal when no proper ideal lies strictly between $\mathfrak{m}$ and the whole ring.

2.3 Quotient criteria

Prime ideals correspond to domain quotients; maximal ideals correspond to field quotients.

2.4 Existence of maximal ideals

Every proper ideal lies in a maximal ideal, using Zorn's lemma.

2.5 Contraction

Inverse images of prime ideals under ring maps are prime when the inverse image remains proper.

2.6 Prime chains

Chains of prime ideals reflect increasing algebraic specialization and later feed into dimension theory.

2.7 Units and maximal ideals

An element is a unit iff it belongs to no maximal ideal.

2.8 Examples

Prime and maximal ideals in $\mathbb{Z}$, polynomial rings, product rings, and localizations illustrate the definitions.

2.9 Core structural results

Theorem 2.1 (Prime quotient criterion). *A proper ideal $\mathfrak{p}$ is prime iff $A/\mathfrak{p}$ is an integral domain.*

Proof. The zero-divisor condition in the quotient is exactly the statement that $ab \in \mathfrak{p}$ forces one factor into $\mathfrak{p}$. $\square$

Theorem 2.2 (Maximal quotient criterion). *A proper ideal $\mathfrak{m}$ is maximal iff $A/\mathfrak{m}$ is a field.*

Proof. Ideals in $A/\mathfrak{m}$ correspond to ideals of A containing $\mathfrak{m}$. A quotient is a field exactly when its only ideals are zero and the whole ring. $\square$

Theorem 2.3 (Maximal implies prime). *Every maximal ideal in a commutative ring with identity is prime.*

Proof. The quotient by a maximal ideal is a field, hence a domain; apply the prime quotient criterion. $\square$

2.10 Worked examples

Example 2.4 (Prime ideals diagnostic). Give a rigorous worked analysis of Prime ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A proper ideal $\mathfrak{p}$ is prime when $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 2.5 (Maximal ideals diagnostic). Give a rigorous worked analysis of Maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A proper ideal $\mathfrak{m}$ is maximal when no proper ideal lies strictly between $\mathfrak{m}$ and the whole ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 2.6 (Quotient criteria diagnostic). Give a rigorous worked analysis of Quotient criteria. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Prime ideals correspond to domain quotients; maximal ideals correspond to field quotients. The decisive step is to use the universal property, exactness

criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 2.7 (Existence of maximal ideals diagnostic). Give a rigorous worked analysis of Existence of maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every proper ideal lies in a maximal ideal, using Zorn's lemma. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

2.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 2.8 (Prime ideals diagnostic). Give a rigorous worked analysis of Prime ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A proper ideal $\mathfrak{p}$ is prime when $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 2.9 (Maximal ideals diagnostic). Give a rigorous worked analysis of Maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A proper ideal $\mathfrak{m}$ is maximal when no proper ideal lies strictly between $\mathfrak{m}$ and the whole ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 2.10 (Quotient criteria diagnostic). Give a rigorous worked analysis of Quotient criteria. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Prime ideals correspond to domain quotients; maximal ideals correspond to field quotients. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 2.11 (Existence of maximal ideals diagnostic). Give a rigorous worked analysis of Existence of maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every proper ideal lies in a maximal ideal, using Zorn's lemma. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 2.12 (Contraction diagnostic). Give a rigorous worked analysis of Contraction. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Inverse images of prime ideals under ring maps are prime when the inverse image remains proper. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 2.13 (Prime chains diagnostic). Give a rigorous worked analysis of Prime chains. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Chains of prime ideals reflect increasing algebraic specialization and later feed into dimension theory. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 2.14 (Units and maximal ideals diagnostic). Give a rigorous worked analysis of Units and maximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element is a unit iff it belongs to no maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 2.15 (Examples diagnostic). Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Prime and maximal ideals in $\mathbb{Z}$, polynomial rings, product rings, and localizations illustrate the definitions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 2.16 (Prime quotient criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A proper ideal $\mathfrak{p}$ is prime iff $A/\mathfrak{p}$ is an integral domain.

Solution. The zero-divisor condition in the quotient is exactly the statement that $ab \in \mathfrak{p}$ forces one factor into $\mathfrak{p}$. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 2.17 (Maximal quotient criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A proper ideal $\mathfrak{m}$ is maximal iff $A/\mathfrak{m}$ is a field.

Solution. Ideals in $A/\mathfrak{m}$ correspond to ideals of A containing $\mathfrak{m}$. A quotient is a field exactly when its only ideals are zero and the whole ring. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 2.18 (Maximal implies prime proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every maximal ideal in a commutative ring with identity is prime.

Solution. The quotient by a maximal ideal is a field, hence a domain; apply the prime quotient criterion. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 2.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Prime and maximal ideals encode the irreducible quotient structures of a commutative ring. Prime ideals detect domains, maximal ideals detect fields, and maximal ideals are automatically prime. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

2.12 Exercises with complete solutions

Exercise 2.20. State and apply the central principle behind Prime ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A proper ideal $\mathfrak{p}$ is prime when $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 2.21. State and apply the central principle behind Maximal ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A proper ideal $\mathfrak{m}$ is maximal when no proper ideal lies strictly between $\mathfrak{m}$ and the whole ring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 2.22. State and apply the central principle behind Quotient criteria. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Prime ideals correspond to domain quotients; maximal ideals correspond to field quotients. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 2.23. State and apply the central principle behind Existence of maximal ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Every proper ideal lies in a maximal ideal, using Zorn's lemma. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 2.24. State and apply the central principle behind Contraction. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Inverse images of prime ideals under ring maps are prime when the inverse image remains proper. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 2.25. State and apply the central principle behind Prime chains. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Chains of prime ideals reflect increasing algebraic specialization and later feed into dimension theory. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 2.26. State and apply the central principle behind Units and maximal ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An element is a unit iff it belongs to no maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 2.27. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Prime and maximal ideals in $\mathbb{Z}$, polynomial rings, product rings, and localizations illustrate the definitions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 3

Radicals and Nilpotents

Radicals and nilpotents measure how equations can vanish after taking powers. The radical of an ideal forgets multiplicity-like information and retains the underlying prime behavior.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

3.1 Nilpotent elements

An element a is nilpotent if $a^n = 0$ for some positive integer n .

3.2 Nilradical

The nilradical $\sqrt{(0)}$ is the ideal of all nilpotent elements.

3.3 Radical of an ideal

$\sqrt{I} = \{a : a^n \in I \text{ for some } n \geq 1\}$.

3.4 Radical ideals

An ideal is radical when $a^n \in I$ implies $a \in I$.

3.5 Quotient interpretation

$\sqrt{I}/I$ is the nilradical of A/I .

3.6 Prime containment

The radical of an ideal is the intersection of all prime ideals containing it.

3.7 Reduced rings

A ring is reduced when its nilradical is zero.

3.8 Examples

Powers of principal ideals, square-zero extensions, and polynomial quotients show how radicals discard nilpotent thickening.

3.9 Core structural results

Theorem 3.1 (Radical is an ideal). *For every ideal I , the set $\sqrt{I}$ is an ideal containing I .*

Proof. Closure under multiplication is immediate. If $a^m, b^n \in I$, then every term in $(a+b)^{m+n}$ contains either at least m copies of a or at least n copies of b , hence lies in I . $\square$

Theorem 3.2 (Quotient nilradical identity). *The nilradical of A/I is $\sqrt{I}/I$.*

Proof. The class of a is nilpotent modulo I exactly when some power a^n lies in I . $\square$

Theorem 3.3 (Radical as intersection of primes). *$\sqrt{I}$ equals the intersection of all prime ideals containing I .*

Proof. One inclusion follows because primes are radical. For the reverse, if $a \notin \sqrt{I}$, localize the quotient A/I at powers of a and choose a maximal ideal there; its contraction is a prime containing I but avoiding a . $\square$

3.10 Worked examples

Example 3.4 (Nilpotent elements diagnostic). Give a rigorous worked analysis of Nilpotent elements. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An element a is nilpotent if $a^n = 0$ for some positive integer n . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 3.5 (Nilradical diagnostic). Give a rigorous worked analysis of Nilradical. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The nilradical $\sqrt{(0)}$ is the ideal of all nilpotent elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 3.6 (Radical of an ideal diagnostic). Give a rigorous worked analysis of Radical of an ideal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\sqrt{I} = \{a : a^n \in I \text{ for some } n \geq 1\}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 3.7 (Radical ideals diagnostic). Give a rigorous worked analysis of Radical ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An ideal is radical when $a^n \in I$ implies $a \in I$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

3.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 3.8 (Nilpotent elements diagnostic). Give a rigorous worked analysis of Nilpotent elements. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element a is nilpotent if $a^n = 0$ for some positive integer n . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 3.9 (Nilradical diagnostic). Give a rigorous worked analysis of Nilradical. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The nilradical $\sqrt{(0)}$ is the ideal of all nilpotent elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 3.10 (Radical of an ideal diagnostic). Give a rigorous worked analysis of Radical of an ideal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\sqrt{I} = \{a : a^n \in I \text{ for some } n \geq 1\}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 3.11 (Radical ideals diagnostic). Give a rigorous worked analysis of Radical ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An ideal is radical when $a^n \in I$ implies $a \in I$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 3.12 (Quotient interpretation diagnostic). Give a rigorous worked analysis of Quotient interpretation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\sqrt{I}/I$ is the nilradical of A/I . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 3.13 (Prime containment diagnostic). Give a rigorous worked analysis of Prime containment. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The radical of an ideal is the intersection of all prime ideals containing it. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 3.14 (Reduced rings diagnostic). Give a rigorous worked analysis of Reduced rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A ring is reduced when its nilradical is zero. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 3.15 (Examples diagnostic). Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Powers of principal ideals, square-zero extensions, and polynomial quotients show how radicals discard nilpotent thickening. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 3.16 (Radical is an ideal proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For every ideal I , the set $\sqrt{I}$ is an ideal containing I .

Solution. Closure under multiplication is immediate. If $a^m, b^n \in I$, then every term in $(a + b)^{m+n}$ contains either at least m copies of a or at least n copies of b , hence lies in I . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 3.17 (Quotient nilradical identity proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: The nilradical of A/I is $\sqrt{I}/I$.

Solution. The class of a is nilpotent modulo I exactly when some power a^n lies in I . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 3.18 (Radical as intersection of primes proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: $\sqrt{I}$ equals the intersection of all prime ideals containing I .

Solution. One inclusion follows because primes are radical. For the reverse, if $a \notin \sqrt{I}$, localize the quotient A/I at powers of a and choose a maximal ideal there; its contraction is a prime containing I but avoiding a . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 3.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Radicals and nilpotents measure how equations can vanish after taking powers. The radical of an ideal forgets multiplicity-like information and retains the underlying prime behavior. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

3.12 Exercises with complete solutions

Exercise 3.20. State and apply the central principle behind Nilpotent elements. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An element a is nilpotent if $a^n = 0$ for some positive integer n . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 3.21. State and apply the central principle behind Nilradical. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The nilradical $\sqrt{(0)}$ is the ideal of all nilpotent elements. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 3.22. State and apply the central principle behind Radical of an ideal. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\sqrt{I} = \{a : a^n \in I \text{ for some } n \geq 1\}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 3.23. State and apply the central principle behind Radical ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An ideal is radical when $a^n \in I$ implies $a \in I$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 3.24. State and apply the central principle behind Quotient interpretation. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\sqrt{I}/I$ is the nilradical of A/I . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 3.25. State and apply the central principle behind Prime containment. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The radical of an ideal is the intersection of all prime ideals containing it. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 3.26. State and apply the central principle behind Reduced rings. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A ring is reduced when its nilradical is zero. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 3.27. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Powers of principal ideals, square-zero extensions, and polynomial quotients show how radicals discard nilpotent thickening. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 4

Chinese Remainder Theory

Chinese remainder theory decomposes rings modulo comaximal ideals into products. It is both a structural theorem and an explicit congruence-solving method.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

4.1 Comaximal ideals

Ideals I, J are comaximal when $I + J = A$.

4.2 Two-ideal CRT

For comaximal I, J , the quotient by $I \cap J$ maps isomorphically to $A/I \times A/J$.

4.3 Product equals intersection

For comaximal ideals, $I \cap J = IJ$.

4.4 Finite families

Pairwise comaximal finite families give product decompositions modulo the intersection or product.

4.5 Explicit idempotents

CRT decompositions produce orthogonal idempotents selecting the different factors.

4.6 Integer congruences

The classical integer CRT is the special case of principal ideals in $\mathbb{Z}$.

4.7 Polynomial interpolation form

Comaximal polynomial ideals encode simultaneous residue conditions.

4.8 Decomposition philosophy

Product rings are algebraic manifestations of independent components.

4.9 Core structural results

Theorem 4.1 (Two-ideal Chinese remainder theorem). *If $I + J = A$, then $A/(I \cap J) \cong A/I \times A/J$.*

Proof. The natural map has kernel $I \cap J$. For surjectivity choose $u \in I, v \in J$ with $u + v = 1$; the element $bv + au$ has prescribed residues a modulo I and b modulo J . $\square$

Theorem 4.2 (Intersection-product equality). *If $I + J = A$, then $I \cap J = IJ$.*

Proof. The inclusion $IJ \subseteq I \cap J$ is automatic. If $x \in I \cap J$ and $1 = u + v$ with $u \in I, v \in J$, then $x = xu + xv$ lies in IJ . $\square$

Theorem 4.3 (Finite CRT). *If $I_1, \dots, I_n$ are pairwise comaximal, then $A/(I_1 \cdots I_n) \cong \prod_i A/I_i$.*

Proof. Induct on n , using that the product of the first $n - 1$ ideals is comaximal with I_n , then apply the two-ideal theorem and the product-intersection identity. $\square$

4.10 Worked examples

Example 4.4 (Comaximal ideals diagnostic). Give a rigorous worked analysis of Comaximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Ideals I, J are comaximal when $I + J = A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 4.5 (Two-ideal CRT diagnostic). Give a rigorous worked analysis of Two-ideal CRT. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For comaximal I, J , the quotient by $I \cap J$ maps isomorphically to $A/I \times A/J$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 4.6 (Product equals intersection diagnostic). Give a rigorous worked analysis of Product equals intersection. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For comaximal ideals, $I \cap J = IJ$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 4.7 (Finite families diagnostic). Give a rigorous worked analysis of Finite families. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Pairwise comaximal finite families give product decompositions modulo the intersection or product. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

4.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 4.8 (Comaximal ideals diagnostic). Give a rigorous worked analysis of Comaximal ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ideals I, J are comaximal when $I + J = A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 4.9 (Two-ideal CRT diagnostic). Give a rigorous worked analysis of Two-ideal CRT. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For comaximal I, J , the quotient by $I \cap J$ maps isomorphically to $A/I \times A/J$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 4.10 (Product equals intersection diagnostic). Give a rigorous worked analysis of Product equals intersection. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For comaximal ideals, $I \cap J = IJ$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 4.11 (Finite families diagnostic). Give a rigorous worked analysis of Finite families. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Pairwise comaximal finite families give product decompositions modulo the intersection or product. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 4.12 (Explicit idempotents diagnostic). Give a rigorous worked analysis of Explicit idempotents. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. CRT decompositions produce orthogonal idempotents selecting the different factors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 4.13 (Integer congruences diagnostic). Give a rigorous worked analysis of Integer congruences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The classical integer CRT is the special case of principal ideals in $\mathbb{Z}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 4.14 (Polynomial interpolation form diagnostic). Give a rigorous worked analysis of Polynomial interpolation form. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Comaximal polynomial ideals encode simultaneous residue conditions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 4.15 (Decomposition philosophy diagnostic). Give a rigorous worked analysis of Decomposition philosophy. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Product rings are algebraic manifestations of independent components. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 4.16 (Two-ideal Chinese remainder theorem proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $I + J = A$, then $A/(I \cap J) \cong A/I \times A/J$.

Solution. The natural map has kernel $I \cap J$. For surjectivity choose $u \in I, v \in J$ with $u + v = 1$; the element $bv + au$ has prescribed residues a modulo I and b modulo J . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 4.17 (Intersection-product equality proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $I + J = A$, then $I \cap J = IJ$.

Solution. The inclusion $IJ \subseteq I \cap J$ is automatic. If $x \in I \cap J$ and $1 = u + v$ with $u \in I, v \in J$, then $x = xu + xv$ lies in IJ . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 4.18 (Finite CRT proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $I_1, \dots, I_n$ are pairwise comaximal, then $A/(I_1 \cdots I_n) \cong \prod_i A/I_i$.

Solution. Induct on n , using that the product of the first $n - 1$ ideals is comaximal with I_n , then apply the two-ideal theorem and the product-intersection identity. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 4.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Chinese remainder theory decomposes rings modulo comaximal ideals into products. It is both a structural theorem and an explicit congruence-solving method. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

4.12 Exercises with complete solutions

Exercise 4.20. State and apply the central principle behind Comaximal ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Ideals I, J are comaximal when $I + J = A$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 4.21. State and apply the central principle behind Two-ideal CRT. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For comaximal I, J , the quotient by $I \cap J$ maps isomorphically to $A/I \times A/J$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 4.22. State and apply the central principle behind Product equals intersection. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For comaximal ideals, $I \cap J = IJ$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 4.23. State and apply the central principle behind Finite families. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Pairwise comaximal finite families give product decompositions modulo the intersection or product. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 4.24. State and apply the central principle behind Explicit idempotents. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. CRT decompositions produce orthogonal idempotents selecting the different factors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 4.25. State and apply the central principle behind Integer congruences. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The classical integer CRT is the special case of principal ideals in $\mathbb{Z}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 4.26. State and apply the central principle behind Polynomial interpolation form. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Comaximal polynomial ideals encode simultaneous residue conditions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 4.27. State and apply the central principle behind Decomposition philosophy. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Product rings are algebraic manifestations of independent components. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Part II

Localization

Chapter 5

Multiplicative Systems

Multiplicative systems specify which elements are to become invertible. They are the input data for localization and control which ideals, primes, and module elements survive.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

5.1 Multiplicative subsets

A multiplicative set S contains 1 and is closed under products; usually $0 \notin S$ for nontrivial localization.

5.2 Saturation

The saturation of S consists of elements that become invertible whenever all of S does.

5.3 Examples

Powers of one element, complements of prime ideals, and complements of unions of primes are standard multiplicative sets.

5.4 Disjointness from ideals

An ideal disjoint from S can survive as a proper ideal after localization.

5.5 Prime avoidance viewpoint

Maximal ideals among ideals disjoint from S are prime.

5.6 Denominator calculus

Fractions identify numerators after multiplying by suitable elements of S .

5.7 Universal goal

Localization is characterized by making S invertible with no extra relations beyond those forced.

5.8 Geometric preview

Choosing $S = \{1, f, f^2, \dots\}$ corresponds to restricting attention to where f is invertible.

5.9 Core structural results

Theorem 5.1 (Maximal disjoint ideal is prime). *If I is maximal among ideals disjoint from a multiplicative set S , then I is prime.*

Proof. If $ab \in I$ with $a, b \notin I$, then both $I + (a)$ and $I + (b)$ meet S . Multiplying corresponding elements of S yields an element of $I + (ab) \subseteq I$, contradicting disjointness. $\square$

Theorem 5.2 (Complement of prime is multiplicative). *If $\mathfrak{p}$ is prime, then $A \setminus \mathfrak{p}$ is multiplicatively closed.*

Proof. If $a, b \notin \mathfrak{p}$ but $ab \in \mathfrak{p}$, primality would force one factor into $\mathfrak{p}$. $\square$

Theorem 5.3 (Powers form a multiplicative set). *For any $f \in A$, $S = \{1, f, f^2, \dots\}$ is multiplicative.*

Proof. The product $f^m f^n = f^{m+n}$ remains in the set and $1 = f^0$ is included. $\square$

5.10 Worked examples

Example 5.4 (Multiplicative subsets diagnostic). Give a rigorous worked analysis of Multiplicative subsets. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A multiplicative set S contains 1 and is closed under products; usually $0 \notin S$ for nontrivial localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 5.5 (Saturation diagnostic). Give a rigorous worked analysis of Saturation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The saturation of S consists of elements that become invertible whenever all of S does. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 5.6 (Examples diagnostic). Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Powers of one element, complements of prime ideals, and complements of unions of primes are standard multiplicative sets. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 5.7 (Disjointness from ideals diagnostic). Give a rigorous worked analysis of Disjointness from ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An ideal disjoint from S can survive as a proper ideal after localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

5.11 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 5.8 (Multiplicative subsets diagnostic). Give a rigorous worked analysis of Multiplicative subsets. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A multiplicative set S contains 1 and is closed under products; usually $0 \notin S$ for nontrivial localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 5.9 (Saturation diagnostic). Give a rigorous worked analysis of Saturation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The saturation of S consists of elements that become invertible whenever all of S does. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 5.10 (Examples diagnostic). Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Powers of one element, complements of prime ideals, and complements of unions of primes are standard multiplicative sets. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 5.11 (Disjointness from ideals diagnostic). Give a rigorous worked analysis of Disjointness from ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An ideal disjoint from S can survive as a proper ideal after localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 5.12 (Prime avoidance viewpoint diagnostic). Give a rigorous worked analysis of Prime avoidance viewpoint. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Maximal ideals among ideals disjoint from S are prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 5.13 (Denominator calculus diagnostic). Give a rigorous worked analysis of Denominator calculus. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Fractions identify numerators after multiplying by suitable elements of S . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 5.14 (Universal goal diagnostic). Give a rigorous worked analysis of Universal goal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization is characterized by making S invertible with no extra relations beyond those forced. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 5.15 (Geometric preview diagnostic). Give a rigorous worked analysis of Geometric preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Choosing $S = \{1, f, f^2, \dots\}$ corresponds to restricting attention to where f is invertible. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 5.16 (Maximal disjoint ideal is prime proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If I is maximal among ideals disjoint from a multiplicative set S , then I is prime.

Solution. If $ab \in I$ with $a, b \notin I$, then both $I + (a)$ and $I + (b)$ meet S . Multiplying corresponding elements of S yields an element of $I + (ab) \subseteq I$, contradicting disjointness. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 5.17 (Complement of prime is multiplicative proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $\mathfrak{p}$ is prime, then $A \setminus \mathfrak{p}$ is multiplicatively closed.

Solution. If $a, b \notin \mathfrak{p}$ but $ab \in \mathfrak{p}$, primality would force one factor into $\mathfrak{p}$. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 5.18 (Powers form a multiplicative set proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For any $f \in A$, $S = \{1, f, f^2, \dots\}$ is multiplicative.

Solution. The product $f^m f^n = f^{m+n}$ remains in the set and $1 = f^0$ is included. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 5.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Multiplicative systems specify which elements are to become invertible. They are the input data for localization and control which ideals, primes, and module elements survive. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

5.12 Exercises with complete solutions

Exercise 5.20. State and apply the central principle behind Multiplicative subsets. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A multiplicative set S contains 1 and is closed under products; usually $0 \notin S$ for nontrivial localization. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 5.21. State and apply the central principle behind Saturation. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The saturation of S consists of elements that become invertible whenever all of S does. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 5.22. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Powers of one element, complements of prime ideals, and complements of unions of primes are standard multiplicative sets. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 5.23. State and apply the central principle behind Disjointness from ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An ideal disjoint from S can survive as a proper ideal after localization. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 5.24. State and apply the central principle behind Prime avoidance viewpoint. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Maximal ideals among ideals disjoint from S are prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 5.25. State and apply the central principle behind Denominator calculus. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Fractions identify numerators after multiplying by suitable elements of S . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 5.26. State and apply the central principle behind Universal goal. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Localization is characterized by making S invertible with no extra relations beyond those forced. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 5.27. State and apply the central principle behind Geometric preview. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Choosing $S = \{1, f, f^2, \dots\}$ corresponds to restricting attention to where f is invertible. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 6

Localization of Rings

Localization of rings adjoins inverses to a chosen multiplicative system. It is governed by a universal property and induces a precise correspondence between primes of the localization and primes of the original ring disjoint from the denominators.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

6.1 Fraction construction

Elements of $S^{-1}A$ are fractions a/s modulo the relation generated by cross-multiplication after multiplying by a further denominator.

6.2 Ring operations

Fractions add and multiply by the usual formulas.

6.3 Canonical map

The map $A \rightarrow S^{-1}A$ sends a to $a/1$ and makes every element of S invertible.

6.4 Universal property

Any ring map from A sending S to units factors uniquely through $S^{-1}A$.

6.5 Zero criterion

A fraction a/s is zero iff some $t \in S$ annihilates a .

6.6 Localized ideals

Extension sends I to $S^{-1}I$, while contraction returns ideals saturated with respect to S .

6.7 Prime correspondence

Prime ideals of $S^{-1}A$ correspond to prime ideals of A disjoint from S .

6.8 Localization at an element

A_f is obtained by inverting powers of one element f .

6.9 Localization at a prime

$A_{\mathfrak{p}}$ inverts everything outside $\mathfrak{p}$ and is a local ring.

6.10 Core structural results

Theorem 6.1 (Universal property of localization). *If $\varphi : A \rightarrow B$ sends every $s \in S$ to a unit, there is a unique map $\tilde{\varphi} : S^{-1}A \rightarrow B$ with $\tilde{\varphi}(a/s) = \varphi(a)\varphi(s)^{-1}$.*

Proof. The formula respects the localization equivalence relation and is forced by the requirement that denominators become inverses, giving existence and uniqueness. $\square$

Theorem 6.2 (Prime correspondence). *Prime ideals of $S^{-1}A$ are in bijection with prime ideals $\mathfrak{p} \subset A$ satisfying $\mathfrak{p} \cap S = \emptyset$.*

Proof. Contraction of a prime avoids units, hence avoids S . For a prime disjoint from S , its extension consists of fractions with numerator in the prime and is prime; extension and contraction are inverse. $\square$

Theorem 6.3 (Localization commutes with quotient). *For an ideal I , $S^{-1}(A/I) \cong S^{-1}A/S^{-1}I$ after replacing S by its image in A/I .*

Proof. Both rings satisfy the same universal property: maps out correspond to maps from A killing I and inverting S . $\square$

6.11 Worked examples

Example 6.4 (Fraction construction diagnostic). Give a rigorous worked analysis of Fraction construction. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Elements of $S^{-1}A$ are fractions a/s modulo the relation generated by cross-multiplication after multiplying by a further denominator. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 6.5 (Ring operations diagnostic). Give a rigorous worked analysis of Ring operations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Fractions add and multiply by the usual formulas. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 6.6 (Canonical map diagnostic). Give a rigorous worked analysis of Canonical map. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The map $A \rightarrow S^{-1}A$ sends a to $a/1$ and makes every element of S invertible. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 6.7 (Universal property diagnostic). Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Any ring map from A sending S to units factors uniquely through $S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

6.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 6.8 (Fraction construction diagnostic). Give a rigorous worked analysis of Fraction construction. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Elements of $S^{-1}A$ are fractions a/s modulo the relation generated by cross-multiplication after multiplying by a further denominator. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 6.9 (Ring operations diagnostic). Give a rigorous worked analysis of Ring operations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Fractions add and multiply by the usual formulas. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 6.10 (Canonical map diagnostic). Give a rigorous worked analysis of Canonical map. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The map $A \rightarrow S^{-1}A$ sends a to $a/1$ and makes every element of S invertible. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 6.11 (Universal property diagnostic). Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Any ring map from A sending S to units factors uniquely through $S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 6.12 (Zero criterion diagnostic). Give a rigorous worked analysis of Zero criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A fraction a/s is zero iff some $t \in S$ annihilates a . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 6.13 (Localized ideals diagnostic). Give a rigorous worked analysis of Localized ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Extension sends I to $S^{-1}I$, while contraction returns ideals saturated with respect to S . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 6.14 (Prime correspondence diagnostic). Give a rigorous worked analysis of Prime correspondence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Prime ideals of $S^{-1}A$ correspond to prime ideals of A disjoint from S . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 6.15 (Localization at an element diagnostic). Give a rigorous worked analysis of Localization at an element. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A_f is obtained by inverting powers of one element f . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 6.16 (Universal property of localization proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $\varphi : A \rightarrow B$ sends every $s \in S$ to a unit, there is a unique map $\tilde{\varphi} : S^{-1}A \rightarrow B$ with $\tilde{\varphi}(a/s) = \varphi(a)\varphi(s)^{-1}$.

Solution. The formula respects the localization equivalence relation and is forced by the requirement that denominators become inverses, giving existence and uniqueness. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 6.17 (Prime correspondence proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Prime ideals of $S^{-1}A$ are in bijection with prime ideals $\mathfrak{p} \subset A$ satisfying $\mathfrak{p} \cap S = \emptyset$.

Solution. Contraction of a prime avoids units, hence avoids S . For a prime disjoint from S , its extension consists of fractions with numerator in the prime and is prime; extension and contraction are inverse. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 6.18 (Localization commutes with quotient proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For an ideal I , $S^{-1}(A/I) \cong S^{-1}A/S^{-1}I$ after replacing S by its image in A/I .

Solution. Both rings satisfy the same universal property: maps out correspond to maps from A killing I and inverting S . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 6.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Localization of rings adjoins inverses to a chosen multiplicative system. It is governed by a universal property and induces a precise correspondence between primes of the localization and primes of the original ring disjoint from the denominators. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

6.13 Exercises with complete solutions

Exercise 6.20. State and apply the central principle behind Fraction construction. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Elements of $S^{-1}A$ are fractions a/s modulo the relation generated by cross-multiplication after multiplying by a further denominator. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 6.21. State and apply the central principle behind Ring operations. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Fractions add and multiply by the usual formulas. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 6.22. State and apply the central principle behind Canonical map. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The map $A \rightarrow S^{-1}A$ sends a to $a/1$ and makes every element of S invertible. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 6.23. State and apply the central principle behind Universal property. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Any ring map from A sending S to units factors uniquely through $S^{-1}A$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 6.24. State and apply the central principle behind Zero criterion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A fraction a/s is zero iff some $t \in S$ annihilates a . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 6.25. State and apply the central principle behind Localized ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Extension sends I to $S^{-1}I$, while contraction returns ideals saturated with respect to S . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 6.26. State and apply the central principle behind Prime correspondence. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Prime ideals of $S^{-1}A$ correspond to prime ideals of A disjoint from S . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 6.27. State and apply the central principle behind Localization at an element. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A_f is obtained by inverting powers of one element f . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 7

Localization of Modules

Localization of modules extends fraction methods from rings to linear algebra over rings. It is exact, commutes with finite algebraic constructions, and detects local behavior of modules.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

7.1 Module fractions

$S^{-1}M$ consists of symbols m/s modulo the expected denominator equivalence.

7.2 Scalar structure

$S^{-1}M$ is naturally a module over $S^{-1}A$.

7.3 Universal property

Maps from M into modules where S acts invertibly factor uniquely through $S^{-1}M$.

7.4 Tensor description

There is a natural isomorphism $S^{-1}M \cong S^{-1}A \otimes_A M$.

7.5 Exactness

Localization preserves exact sequences.

7.6 Submodules and quotients

Localization commutes with quotients and sends submodules to localized submodules.

7.7 Vanishing criterion

An element $m/1$ vanishes locally iff some denominator in S annihilates m .

7.8 Finite generation

Localization preserves finite generation and finite presentation.

7.9 Local detection

Many module properties can be tested after localizing at all prime or maximal ideals.

7.10 Core structural results

Theorem 7.1 (Tensor description of localization). *There is a natural isomorphism $S^{-1}A \otimes_A M \cong S^{-1}M$ sending $(a/s) \otimes m$ to am/s .*

Proof. The formula is balanced and induces a homomorphism. Its inverse sends m/s to $(1/s) \otimes m$; the two composites are identities. $\square$

Theorem 7.2 (Exactness of localization). *If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact, then $0 \rightarrow S^{-1}M' \rightarrow S^{-1}M \rightarrow S^{-1}M'' \rightarrow 0$ is exact.*

Proof. Surjectivity is immediate on fractions. For injectivity or kernel equality, if a localized element maps to zero, some denominator annihilates the relevant numerator, allowing the numerator to be moved into the preceding module before localization. $\square$

Theorem 7.3 (Localization commutes with quotient). *For $N \subset M$, $S^{-1}(M/N) \cong S^{-1}M/S^{-1}N$.*

Proof. Apply exactness to $0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$. $\square$

7.11 Worked examples

Example 7.4 (Module fractions diagnostic). Give a rigorous worked analysis of Module fractions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $S^{-1}M$ consists of symbols m/s modulo the expected denominator equivalence. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 7.5 (Scalar structure diagnostic). Give a rigorous worked analysis of Scalar structure. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $S^{-1}M$ is naturally a module over $S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 7.6 (Universal property diagnostic). Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Maps from M into modules where S acts invertibly factor uniquely through $S^{-1}M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 7.7 (Tensor description diagnostic). Give a rigorous worked analysis of Tensor description. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. There is a natural isomorphism $S^{-1}M \cong S^{-1}A \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

7.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 7.8 (Module fractions diagnostic). Give a rigorous worked analysis of Module fractions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $S^{-1}M$ consists of symbols m/s modulo the expected denominator equivalence. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 7.9 (Scalar structure diagnostic). Give a rigorous worked analysis of Scalar structure. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $S^{-1}M$ is naturally a module over $S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 7.10 (Universal property diagnostic). Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Maps from M into modules where S acts invertibly factor uniquely through $S^{-1}M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 7.11 (Tensor description diagnostic). Give a rigorous worked analysis of Tensor description. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. There is a natural isomorphism $S^{-1}M \cong S^{-1}A \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 7.12 (Exactness diagnostic). Give a rigorous worked analysis of Exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization preserves exact sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 7.13 (Submodules and quotients diagnostic). Give a rigorous worked analysis of Submodules and quotients. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization commutes with quotients and sends submodules to localized submodules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 7.14 (Vanishing criterion diagnostic). Give a rigorous worked analysis of Vanishing criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element $m/1$ vanishes locally iff some denominator in S annihilates m . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 7.15 (Finite generation diagnostic). Give a rigorous worked analysis of Finite generation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization preserves finite generation and finite presentation. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 7.16 (Tensor description of localization proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: There is a natural isomorphism $S^{-1}A \otimes_A M \cong S^{-1}M$ sending $(a/s) \otimes m$ to am/s .

Solution. The formula is balanced and induces a homomorphism. Its inverse sends m/s to $(1/s) \otimes m$; the two composites are identities. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 7.17 (Exactness of localization proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact, then $0 \rightarrow S^{-1}M' \rightarrow S^{-1}M \rightarrow S^{-1}M'' \rightarrow 0$ is exact.

Solution. Surjectivity is immediate on fractions. For injectivity or kernel equality, if a localized element maps to zero, some denominator annihilates the relevant numerator, allowing the numerator to be moved into the preceding module before localization. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 7.18 (Localization commutes with quotient proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $N \subset M$, $S^{-1}(M/N) \cong S^{-1}M/S^{-1}N$.

Solution. Apply exactness to $0 \rightarrow N \rightarrow M \rightarrow M/N \rightarrow 0$. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 7.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Localization of modules extends fraction methods from rings to linear algebra over rings. It is exact, commutes with finite algebraic constructions, and detects local behavior of modules. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

7.13 Exercises with complete solutions

Exercise 7.20. State and apply the central principle behind Module fractions. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $S^{-1}M$ consists of symbols m/s modulo the expected denominator equivalence. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 7.21. State and apply the central principle behind Scalar structure. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $S^{-1}M$ is naturally a module over $S^{-1}A$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 7.22. State and apply the central principle behind Universal property. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Maps from M into modules where S acts invertibly factor uniquely through $S^{-1}M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 7.23. State and apply the central principle behind Tensor description. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. There is a natural isomorphism $S^{-1}M \cong S^{-1}A \otimes_A M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 7.24. State and apply the central principle behind Exactness. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Localization preserves exact sequences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 7.25. State and apply the central principle behind Submodules and quotients. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Localization commutes with quotients and sends submodules to localized submodules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 7.26. State and apply the central principle behind Vanishing criterion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An element $m/1$ vanishes locally iff some denominator in S annihilates m . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 7.27. State and apply the central principle behind Finite generation. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Localization preserves finite generation and finite presentation. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 8

Local Rings and Localization at Primes

Local rings concentrate algebra around one maximal ideal. Localization at a prime produces the canonical local ring attached to that prime, and unit tests become especially simple.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

8.1 Local rings

A local ring has exactly one maximal ideal.

8.2 Nonunits

In a local ring, the nonunits form the unique maximal ideal.

8.3 Unit criterion

An element is a unit precisely when it lies outside the maximal ideal.

8.4 Localization at primes

For a prime $\mathfrak{p}$, $A_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.

8.5 Residue field

The field $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is the fraction field of $A/\mathfrak{p}$ with respect to nonzero classes.

8.6 Nakayama preview

Local rings make finite-generation arguments rigid because the maximal ideal controls reduction modulo the residue field.

8.7 Local homomorphisms

A homomorphism of local rings is local when the maximal ideal pulls back to the maximal ideal.

8.8 Examples

Discrete valuation rings, localized polynomial rings, and power-series rings provide standard local examples.

8.9 Local-global philosophy

Properties of modules and ideals are often checked prime by prime using these local rings.

8.10 Core structural results

Theorem 8.1 (Nonunits characterize local rings). *A commutative ring is local iff its nonunits form an ideal; in that case this ideal is the unique maximal ideal.*

Proof. Every proper ideal consists of nonunits. If nonunits form an ideal, it is proper and contains every proper ideal, hence is uniquely maximal. Conversely, in a local ring every element outside the maximal ideal generates the unit ideal and is therefore a unit. $\square$

Theorem 8.2 (Prime localization is local). *For a prime $\mathfrak{p}$, the ring $A_{\mathfrak{p}}$ is local with unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.*

Proof. Prime ideals of the localization correspond to primes of A contained in $\mathfrak{p}$. The extension of $\mathfrak{p}$ contains every such prime and is therefore the unique maximal ideal. $\square$

Theorem 8.3 (Residue field identification). *The residue field of $A_{\mathfrak{p}}$ is canonically isomorphic to the fraction field of $A/\mathfrak{p}$.*

Proof. Modulo $\mathfrak{p}A_{\mathfrak{p}}$, numerators and denominators descend to $A/\mathfrak{p}$, and every denominator outside $\mathfrak{p}$ becomes a nonzero class, hence invertible in the fraction field. $\square$

8.11 Worked examples

Example 8.4 (Local rings diagnostic). Give a rigorous worked analysis of Local rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A local ring has exactly one maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 8.5 (Nonunits diagnostic). Give a rigorous worked analysis of Nonunits. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. In a local ring, the nonunits form the unique maximal ideal. The decisive step

is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 8.6 (Unit criterion diagnostic). Give a rigorous worked analysis of Unit criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An element is a unit precisely when it lies outside the maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 8.7 (Localization at primes diagnostic). Give a rigorous worked analysis of Localization at primes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For a prime $\mathfrak{p}$, $A_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

8.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 8.8 (Local rings diagnostic). Give a rigorous worked analysis of Local rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A local ring has exactly one maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 8.9 (Nonunits diagnostic). Give a rigorous worked analysis of Nonunits. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. In a local ring, the nonunits form the unique maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 8.10 (Unit criterion diagnostic). Give a rigorous worked analysis of Unit criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element is a unit precisely when it lies outside the maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 8.11 (Localization at primes diagnostic). Give a rigorous worked analysis of Localization at primes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For a prime $\mathfrak{p}$, $A_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 8.12 (Residue field diagnostic). Give a rigorous worked analysis of Residue field. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The field $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is the fraction field of $A/\mathfrak{p}$ with respect to nonzero classes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 8.13 (Nakayama preview diagnostic). Give a rigorous worked analysis of Nakayama preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Local rings make finite-generation arguments rigid because the maximal ideal controls reduction modulo the residue field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 8.14 (Local homomorphisms diagnostic). Give a rigorous worked analysis of Local homomorphisms. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A homomorphism of local rings is local when the maximal ideal pulls back to the maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 8.15 (Examples diagnostic). Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Discrete valuation rings, localized polynomial rings, and power-series rings provide standard local examples. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 8.16 (Nonunits characterize local rings proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A commutative ring is local iff its nonunits form an ideal; in that case this ideal is the unique maximal ideal.

Solution. Every proper ideal consists of nonunits. If nonunits form an ideal, it is proper and contains every proper ideal, hence is uniquely maximal. Conversely, in a local ring every element outside the maximal ideal generates the unit ideal and is therefore a unit. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 8.17 (Prime localization is local proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For a prime $\mathfrak{p}$, the ring $A_{\mathfrak{p}}$ is local with unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.

Solution. Prime ideals of the localization correspond to primes of A contained in $\mathfrak{p}$. The extension of $\mathfrak{p}$ contains every such prime and is therefore the unique maximal ideal. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 8.18 (Residue field identification proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: The residue field of $A_{\mathfrak{p}}$ is canonically isomorphic to the fraction field of $A/\mathfrak{p}$.

Solution. Modulo $\mathfrak{p}A_{\mathfrak{p}}$, numerators and denominators descend to $A/\mathfrak{p}$, and every denominator outside $\mathfrak{p}$ becomes a nonzero class, hence invertible in the fraction field. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 8.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Local rings concentrate algebra around one maximal ideal. Localization at a prime produces the canonical local ring attached to that prime, and unit tests become especially simple. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

8.13 Exercises with complete solutions

Exercise 8.20. State and apply the central principle behind Local rings. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A local ring has exactly one maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 8.21. State and apply the central principle behind Nonunits. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. In a local ring, the nonunits form the unique maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 8.22. State and apply the central principle behind Unit criterion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An element is a unit precisely when it lies outside the maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 8.23. State and apply the central principle behind Localization at primes. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For a prime $\mathfrak{p}$, $A_{\mathfrak{p}}$ has maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 8.24. State and apply the central principle behind Residue field. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The field $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is the fraction field of $A/\mathfrak{p}$ with respect to nonzero classes. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 8.25. State and apply the central principle behind Nakayama preview. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Local rings make finite-generation arguments rigid because the maximal ideal controls reduction modulo the residue field. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 8.26. State and apply the central principle behind Local homomorphisms. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A homomorphism of local rings is local when the maximal ideal pulls back to the maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 8.27. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Discrete valuation rings, localized polynomial rings, and power-series rings provide standard local examples. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Part III

Modules and Tensor Products

Chapter 9

Modules and Exact Sequences

Modules generalize vector spaces by allowing scalars from an arbitrary ring. Exact sequences record how modules are assembled from submodules and quotients and become the basic grammar of homological algebra.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

9.1 Modules

An A -module is an abelian group with compatible scalar multiplication by A .

9.2 Submodules and quotients

Submodules are additive subgroups stable under scalars; quotient modules encode congruence modulo a submodule.

9.3 Module maps

Kernels, images, and cokernels are modules and satisfy the familiar first isomorphism theorem.

9.4 Exact sequences

A sequence is exact when the image at each term equals the next kernel.

9.5 Short exact sequences

$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ expresses M' as a submodule and M'' as the corresponding quotient.

9.6 Splittings

A short exact sequence splits when the middle module is isomorphic to a direct sum of its ends.

9.7 Direct sums

Direct sums collect finitely supported tuples and implement coproducts in the module category.

9.8 Diagram arguments

Exact sequences organize kernel-image calculations and support diagram lemmas later in homological algebra.

9.9 Examples

Ideals, quotient modules, free modules, torsion abelian groups, and vector spaces all fit the same language.

9.10 Core structural results

Theorem 9.1 (First isomorphism theorem for modules). *For an A -linear map $f : M \rightarrow N$, $M/\ker f \cong \operatorname{im} f$.*

Proof. Send $m + \ker f$ to $f(m)$. The same kernel argument as for groups and rings gives a well-defined bijective linear map onto the image. $\square$

Theorem 9.2 (Splitting criterion). *A short exact sequence $0 \rightarrow M' \xrightarrow{i} M \xrightarrow{p} M'' \rightarrow 0$ splits iff p has a section, equivalently iff i has a retraction.*

Proof. A section s gives $M = i(M') \oplus s(M'')$. Conversely a direct-sum decomposition supplies both projection and inclusion maps realizing the section and retraction. $\square$

Theorem 9.3 (Exactness of direct sums). *Taking arbitrary direct sums of short exact sequences of modules preserves exactness.*

Proof. All maps act componentwise. An element of a direct sum has finite support, so kernel and image membership can be checked component by component. $\square$

9.11 Worked examples

Example 9.4 (Modules diagnostic). Give a rigorous worked analysis of Modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An A -module is an abelian group with compatible scalar multiplication by A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 9.5 (Submodules and quotients diagnostic). Give a rigorous worked analysis of Submodules and quotients. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Submodules are additive subgroups stable under scalars; quotient modules encode congruence modulo a submodule. The decisive step

is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 9.6 (Module maps diagnostic). Give a rigorous worked analysis of Module maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Kernels, images, and cokernels are modules and satisfy the familiar first isomorphism theorem. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 9.7 (Exact sequences diagnostic). Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A sequence is exact when the image at each term equals the next kernel. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

9.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 9.8 (Modules diagnostic). Give a rigorous worked analysis of Modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An A -module is an abelian group with compatible scalar multiplication by A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 9.9 (Submodules and quotients diagnostic). Give a rigorous worked analysis of Submodules and quotients. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Submodules are additive subgroups stable under scalars; quotient modules encode congruence modulo a submodule. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 9.10 (Module maps diagnostic). Give a rigorous worked analysis of Module maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Kernels, images, and cokernels are modules and satisfy the familiar first isomorphism theorem. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 9.11 (Exact sequences diagnostic). Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A sequence is exact when the image at each term equals the next kernel. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 9.12 (Short exact sequences diagnostic). Give a rigorous worked analysis of Short exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ expresses M' as a submodule and M'' as the corresponding quotient. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 9.13 (Splittings diagnostic). Give a rigorous worked analysis of Splittings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A short exact sequence splits when the middle module is isomorphic to a direct sum of its ends. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 9.14 (Direct sums diagnostic). Give a rigorous worked analysis of Direct sums. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Direct sums collect finitely supported tuples and implement coproducts in the module category. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 9.15 (Diagram arguments diagnostic). Give a rigorous worked analysis of Diagram arguments. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Exact sequences organize kernel-image calculations and support diagram lemmas later in homological algebra. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 9.16 (First isomorphism theorem for modules proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For an A -linear map $f : M \rightarrow N$, $M/\ker f \cong \operatorname{im} f$.

Solution. Send $m + \ker f$ to $f(m)$. The same kernel argument as for groups and rings gives a well-defined bijective linear map onto the image. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 9.17 (Splitting criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A short exact sequence $0 \rightarrow M' \xrightarrow{i} M \xrightarrow{p} M'' \rightarrow 0$ splits iff p has a section, equivalently iff i has a retraction.

Solution. A section s gives $M = i(M') \oplus s(M'')$. Conversely a direct-sum decomposition supplies both projection and inclusion maps realizing the section and retraction. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 9.18 (Exactness of direct sums proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Taking arbitrary direct sums of short exact sequences of modules preserves exactness.

Solution. All maps act componentwise. An element of a direct sum has finite support, so kernel and image membership can be checked component by component. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 9.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Modules generalize vector spaces by allowing scalars from an arbitrary ring. Exact sequences record how modules are assembled from submodules and quotients and become the basic grammar of homological algebra. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

9.13 Exercises with complete solutions

Exercise 9.20. State and apply the central principle behind Modules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An A -module is an abelian group with compatible scalar multiplication by A . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 9.21. State and apply the central principle behind Submodules and quotients. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Submodules are additive subgroups stable under scalars; quotient modules encode congruence modulo a submodule. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 9.22. State and apply the central principle behind Module maps. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Kernels, images, and cokernels are modules and satisfy the familiar first isomorphism theorem. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 9.23. State and apply the central principle behind Exact sequences. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A sequence is exact when the image at each term equals the next kernel. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 9.24. State and apply the central principle behind Short exact sequences. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ expresses M' as a submodule and M'' as the corresponding quotient. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 9.25. State and apply the central principle behind Splittings. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A short exact sequence splits when the middle module is isomorphic to a direct sum of its ends. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 9.26. State and apply the central principle behind Direct sums. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Direct sums collect finitely supported tuples and implement coproducts in the module category. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 9.27. State and apply the central principle behind Diagram arguments. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Exact sequences organize kernel-image calculations and support diagram lemmas later in homological algebra. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 10

Tensor Products

The tensor product converts bilinear maps into linear maps. It controls base change, localization, flatness, and many multilinear constructions throughout algebra and geometry.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

10.1 Balanced bilinear maps

An A -balanced map satisfies $b(am, n) = b(m, an)$ in addition to additivity in each variable.

10.2 Tensor product

$M \otimes_A N$ is generated by symbols $m \otimes n$ modulo additivity and balancing relations.

10.3 Universal property

Balanced bilinear maps out of $M \times N$ correspond uniquely to linear maps out of $M \otimes_A N$.

10.4 Basic identities

$A \otimes_A M \cong M$ and direct sums commute with tensor products.

10.5 Quotient tensors

$(A/I) \otimes_A M \cong M/IM$.

10.6 Associativity and symmetry

Canonical isomorphisms identify iterated tensor products and swap factors over a commutative ring.

10.7 Right exactness

Tensoring preserves cokernels and surjections, but not necessarily injections.

10.8 Localization

$$S^{-1}A \otimes_A M \cong S^{-1}M.$$

10.9 Examples

Cyclic groups, quotient modules, field extensions, and polynomial base change exhibit torsion and dimension phenomena.

10.10 Multilinear extension

Exterior and symmetric powers are built by further quotienting tensor powers.

10.11 Core structural results

Theorem 10.1 (Universal property). *For every balanced bilinear map $b : M \times N \rightarrow P$, there is a unique linear map $\tilde{b} : M \otimes_A N \rightarrow P$ with $\tilde{b}(m \otimes n) = b(m, n)$.*

Proof. Construct the tensor product as the free module on pairs modulo the subgroup generated by additivity and balancing relations. The relations are exactly those killed by every balanced bilinear map. $\square$

Theorem 10.2 (Quotient tensor identity). *For an ideal $I \subset A$, $(A/I) \otimes_A M \cong M/IM$.*

Proof. Send $(a + I) \otimes m$ to $am + IM$. The map is balanced and surjective; the universal properties on both sides show that its kernel is precisely the relations generated by IM . $\square$

Theorem 10.3 (Right exactness of tensor). *If $M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact, then $M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0$ is exact.*

Proof. The tensor product is a left adjoint in either variable, so it preserves cokernels; directly, generators in the kernel of the final map differ by tensors coming from the preceding module. $\square$

10.12 Worked examples

Example 10.4 (Balanced bilinear maps diagnostic). Give a rigorous worked analysis of Balanced bilinear maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An A -balanced map satisfies $b(am, n) = b(m, an)$ in addition to additivity in each variable. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 10.5 (Tensor product diagnostic). Give a rigorous worked analysis of Tensor product. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $M \otimes_A N$ is generated by symbols $m \otimes n$ modulo additivity and balancing relations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 10.6 (Universal property diagnostic). Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Balanced bilinear maps out of $M \times N$ correspond uniquely to linear maps out of $M \otimes_A N$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 10.7 (Basic identities diagnostic). Give a rigorous worked analysis of Basic identities. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $A \otimes_A M \cong M$ and direct sums commute with tensor products. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

10.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 10.8 (Balanced bilinear maps diagnostic). Give a rigorous worked analysis of Balanced bilinear maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An A -balanced map satisfies $b(am, n) = b(m, an)$ in addition to additivity in each variable. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 10.9 (Tensor product diagnostic). Give a rigorous worked analysis of Tensor product. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $M \otimes_A N$ is generated by symbols $m \otimes n$ modulo additivity and balancing relations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 10.10 (Universal property diagnostic). Give a rigorous worked analysis of Universal property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Balanced bilinear maps out of $M \times N$ correspond uniquely to linear maps out of $M \otimes_A N$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 10.11 (Basic identities diagnostic). Give a rigorous worked analysis of Basic identities. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $A \otimes_A M \cong M$ and direct sums commute with tensor products. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 10.12 (Quotient tensors diagnostic). Give a rigorous worked analysis of Quotient tensors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $(A/I) \otimes_A M \cong M/IM$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 10.13 (Associativity and symmetry diagnostic). Give a rigorous worked analysis of Associativity and symmetry. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Canonical isomorphisms identify iterated tensor products and swap factors over a commutative ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 10.14 (Right exactness diagnostic). Give a rigorous worked analysis of Right exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Tensoring preserves cokernels and surjections, but not necessarily injections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 10.15 (Localization diagnostic). Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $S^{-1}A \otimes_A M \cong S^{-1}M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 10.16 (Universal property proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For every balanced bilinear map $b : M \times N \rightarrow P$, there is a unique linear map $\tilde{b} : M \otimes_A N \rightarrow P$ with $\tilde{b}(m \otimes n) = b(m, n)$.

Solution. Construct the tensor product as the free module on pairs modulo the subgroup generated by additivity and balancing relations. The relations are exactly those killed by every balanced bilinear map. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 10.17 (Quotient tensor identity proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For an ideal $I \subset A$, $(A/I) \otimes_A M \cong M/IM$.

Solution. Send $(a + I) \otimes m$ to $am + IM$. The map is balanced and surjective; the universal properties on both sides show that its kernel is precisely the relations generated by IM . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 10.18 (Right exactness of tensor proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact, then $M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0$ is exact.

Solution. The tensor product is a left adjoint in either variable, so it preserves cokernels; directly, generators in the kernel of the final map differ by tensors coming from the preceding module. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 10.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. The tensor product converts bilinear maps into linear maps. It controls base change, localization, flatness, and many multilinear constructions throughout algebra and geometry. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

10.14 Exercises with complete solutions

Exercise 10.20. State and apply the central principle behind Balanced bilinear maps. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An A -balanced map satisfies $b(am, n) = b(m, an)$ in addition to additivity in each variable. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 10.21. State and apply the central principle behind Tensor product. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $M \otimes_A N$ is generated by symbols $m \otimes n$ modulo additivity and balancing relations. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 10.22. State and apply the central principle behind Universal property. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Balanced bilinear maps out of $M \times N$ correspond uniquely to linear maps out of $M \otimes_A N$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 10.23. State and apply the central principle behind Basic identities. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $A \otimes_A M \cong M$ and direct sums commute with tensor products. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 10.24. State and apply the central principle behind Quotient tensors. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $(A/I) \otimes_A M \cong M/IM$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 10.25. State and apply the central principle behind Associativity and symmetry. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Canonical isomorphisms identify iterated tensor products and swap factors over a commutative ring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 10.26. State and apply the central principle behind Right exactness. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Tensoring preserves cokernels and surjections, but not necessarily injections. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 10.27. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $S^{-1}A \otimes_A M \cong S^{-1}M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 11

Quotients and Base Change

Quotients and base change describe how algebraic relations behave after changing the scalar ring. Tensor products are the mechanism that transports modules, presentations, and exact sequences to the new base.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

11.1 Scalar extension

For $A \rightarrow B$, the base-changed module is $B \otimes_A M$.

11.2 Restriction of scalars

Every B -module is an A -module by composing scalar actions with $A \rightarrow B$.

11.3 Quotient base change

$B \otimes_A (A/I)$ is naturally B/IB .

11.4 Submodule caution

Base change need not preserve injections unless the new scalar module is flat.

11.5 Presentations

Base changing a presentation replaces its matrices by the same matrices with entries mapped into B .

11.6 Localization as base change

Localization is scalar extension along $A \rightarrow S^{-1}A$.

11.7 Residue-field fibers

$M \otimes_A \kappa(\mathfrak{p})$ measures the fiber of a module at a prime.

11.8 Iterated base change

For $A \rightarrow B \rightarrow C$, there is a canonical isomorphism $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$.

11.9 Compatibility with quotients

Cokernels commute with base change because tensor is right exact.

11.10 Core structural results

Theorem 11.1 (Quotient under base change). *For $A \rightarrow B$ and ideal $I \subset A$, $B \otimes_A A/I \cong B/IB$.*

Proof. Apply the quotient tensor identity to the A -module B ; the submodule IB is generated by images of elements of I . $\square$

Theorem 11.2 (Transitivity of base change). *For $A \rightarrow B \rightarrow C$ and an A -module M , $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$.*

Proof. Send $c \otimes (b \otimes m)$ to $cb \otimes m$. Balancedness gives a well-defined map, and the inverse sends $c \otimes m$ to $c \otimes (1 \otimes m)$. $\square$

Theorem 11.3 (Base change of a cokernel). *If $M' \rightarrow M \rightarrow Q \rightarrow 0$ is exact, then $B \otimes_A M' \rightarrow B \otimes_A M \rightarrow B \otimes_A Q \rightarrow 0$ is exact.*

Proof. This is the right exactness of tensor product with the A -module B . $\square$

11.11 Worked examples

Example 11.4 (Scalar extension diagnostic). Give a rigorous worked analysis of Scalar extension. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For $A \rightarrow B$, the base-changed module is $B \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 11.5 (Restriction of scalars diagnostic). Give a rigorous worked analysis of Restriction of scalars. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every B -module is an A -module by composing scalar actions with $A \rightarrow B$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 11.6 (Quotient base change diagnostic). Give a rigorous worked analysis of Quotient base change. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $B \otimes_A (A/I)$ is naturally B/IB . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 11.7 (Submodule caution diagnostic). Give a rigorous worked analysis of Submodule caution. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Base change need not preserve injections unless the new scalar module is flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

11.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 11.8 (Scalar extension diagnostic). Give a rigorous worked analysis of Scalar extension. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For $A \rightarrow B$, the base-changed module is $B \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 11.9 (Restriction of scalars diagnostic). Give a rigorous worked analysis of Restriction of scalars. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every B -module is an A -module by composing scalar actions with $A \rightarrow B$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 11.10 (Quotient base change diagnostic). Give a rigorous worked analysis of Quotient base change. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $B \otimes_A (A/I)$ is naturally B/IB . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 11.11 (Submodule caution diagnostic). Give a rigorous worked analysis of Submodule caution. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Base change need not preserve injections unless the new scalar module is flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 11.12 (Presentations diagnostic). Give a rigorous worked analysis of Presentations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Base changing a presentation replaces its matrices by the same matrices with entries mapped into B . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 11.13 (Localization as base change diagnostic). Give a rigorous worked analysis of Localization as base change. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization is scalar extension along $A \rightarrow S^{-1}A$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 11.14 (Residue-field fibers diagnostic). Give a rigorous worked analysis of Residue-field fibers. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $M \otimes_A \kappa(\mathfrak{p})$ measures the fiber of a module at a prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 11.15 (Iterated base change diagnostic). Give a rigorous worked analysis of Iterated base change. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For $A \rightarrow B \rightarrow C$, there is a canonical isomorphism $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 11.16 (Quotient under base change proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $A \rightarrow B$ and ideal $I \subset A$, $B \otimes_A A/I \cong B/IB$.

Solution. Apply the quotient tensor identity to the A -module B ; the submodule IB is generated by images of elements of I . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 11.17 (Transitivity of base change proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $A \rightarrow B \rightarrow C$ and an A -module M , $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$.

Solution. Send $c \otimes (b \otimes m)$ to $cb \otimes m$. Balancedness gives a well-defined map, and the inverse sends $c \otimes m$ to $c \otimes (1 \otimes m)$. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 11.18 (Base change of a cokernel proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $M' \rightarrow M \rightarrow Q \rightarrow 0$ is exact, then $B \otimes_A M' \rightarrow B \otimes_A M \rightarrow B \otimes_A Q \rightarrow 0$ is exact.

Solution. This is the right exactness of tensor product with the A -module B . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 11.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Quotients and base change describe how algebraic relations behave after changing the scalar ring. Tensor products are the mechanism that transports modules, presentations, and exact sequences to the new base. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

11.13 Exercises with complete solutions

Exercise 11.20. State and apply the central principle behind Scalar extension. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For $A \rightarrow B$, the base-changed module is $B \otimes_A M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 11.21. State and apply the central principle behind Restriction of scalars. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Every B -module is an A -module by composing scalar actions with $A \rightarrow B$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 11.22. State and apply the central principle behind Quotient base change. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $B \otimes_A (A/I)$ is naturally B/IB . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 11.23. State and apply the central principle behind Submodule caution. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Base change need not preserve injections unless the new scalar module is flat. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 11.24. State and apply the central principle behind Presentations. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Base changing a presentation replaces its matrices by the same matrices with entries mapped into B . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 11.25. State and apply the central principle behind Localization as base change. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Localization is scalar extension along $A \rightarrow S^{-1}A$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 11.26. State and apply the central principle behind Residue-field fibers. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $M \otimes_A \kappa(\mathfrak{p})$ measures the fiber of a module at a prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 11.27. State and apply the central principle behind Iterated base change. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For $A \rightarrow B \rightarrow C$, there is a canonical isomorphism $C \otimes_B (B \otimes_A M) \cong C \otimes_A M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 12

Hom and Finitely Presented Modules

Hom and finite presentation control maps out of modules and the stability of algebraic data under limits and base change. Finitely presented modules are those described by finitely many generators and finitely many relations.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

12.1 Hom modules

For A -modules M, N , $\mathrm{Hom}_A(M, N)$ is an A -module under pointwise operations.

12.2 Contravariance and covariance

Hom is contravariant in its first argument and covariant in its second.

12.3 Left exactness

$\mathrm{Hom}_A(M, -)$ preserves kernels but not generally surjections.

12.4 Finite generation

A module is finitely generated if a finite free module surjects onto it.

12.5 Finite presentation

A module is finitely presented if it is the cokernel of a map between finite free modules.

12.6 Generators and relations

A finite presentation packages finitely many generators together with finitely many module relations.

12.7 Hom and filtered colimits

Finite presentation is characterized by commuting of Hom out of the module with filtered colimits.

12.8 Base change of presentations

Finite presentations remain finite after scalar extension.

12.9 Localization of Hom

For finitely presented M , localization commutes with $\text{Hom}_A(M, N)$.

12.10 Matrix viewpoint

Maps between finite free modules are matrices, so finite presentations translate module questions into matrix algebra.

12.11 Core structural results

Theorem 12.1 (Finite presentation criterion). *M is finitely presented iff there exists an exact sequence $A^m \rightarrow A^n \rightarrow M \rightarrow 0$ with finite m, n .*

Proof. This is the definition translated into generators and relations: the second map chooses finitely many generators and the image of the first supplies finitely many generators for all relations. $\square$

Theorem 12.2 (Hom is left exact). *For $0 \rightarrow N' \rightarrow N \rightarrow N''$, the sequence $0 \rightarrow \text{Hom}(M, N') \rightarrow \text{Hom}(M, N) \rightarrow \text{Hom}(M, N'')$ is exact.*

Proof. A map into N lands in N' exactly when its composite with $N \rightarrow N''$ is zero. Injectivity follows because $N' \rightarrow N$ is injective. $\square$

Theorem 12.3 (Localization of Hom for finitely presented modules). *If M is finitely presented, then $S^{-1}\text{Hom}_A(M, N) \cong \text{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N)$.*

Proof. Choose a finite presentation of M , apply Hom to it, localize the resulting kernel description, and use exactness of localization together with the evident formula for finite free modules. $\square$

12.12 Worked examples

Example 12.4 (Hom modules diagnostic). Give a rigorous worked analysis of Hom modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For A -modules M, N , $\text{Hom}_A(M, N)$ is an A -module under pointwise

operations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 12.5 (Contravariance and covariance diagnostic). Give a rigorous worked analysis of Contravariance and covariance. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Hom is contravariant in its first argument and covariant in its second. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 12.6 (Left exactness diagnostic). Give a rigorous worked analysis of Left exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\text{Hom}_A(M, -)$ preserves kernels but not generally surjections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 12.7 (Finite generation diagnostic). Give a rigorous worked analysis of Finite generation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A module is finitely generated if a finite free module surjects onto it. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

12.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 12.8 (Hom modules diagnostic). Give a rigorous worked analysis of Hom modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For A -modules M, N , $\text{Hom}_A(M, N)$ is an A -module under pointwise operations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 12.9 (Contravariance and covariance diagnostic). Give a rigorous worked analysis of Contravariance and covariance. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Hom is contravariant in its first argument and covariant in its second. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 12.10 (Left exactness diagnostic). Give a rigorous worked analysis of Left exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\text{Hom}_A(M, -)$ preserves kernels but not generally surjections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 12.11 (Finite generation diagnostic). Give a rigorous worked analysis of Finite generation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is finitely generated if a finite free module surjects onto it. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 12.12 (Finite presentation diagnostic). Give a rigorous worked analysis of Finite presentation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is finitely presented if it is the cokernel of a map between finite free modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 12.13 (Generators and relations diagnostic). Give a rigorous worked analysis of Generators and relations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A finite presentation packages finitely many generators together with finitely many module relations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 12.14 (Hom and filtered colimits diagnostic). Give a rigorous worked analysis of Hom and filtered colimits. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite presentation is characterized by commuting of Hom out of the module with filtered colimits. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 12.15 (Base change of presentations diagnostic). Give a rigorous worked analysis of Base change of presentations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite presentations remain finite after scalar extension. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 12.16 (Finite presentation criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: M is finitely presented iff there exists an exact sequence $A^m \rightarrow A^n \rightarrow M \rightarrow 0$ with finite m, n .

Solution. This is the definition translated into generators and relations: the second map chooses finitely many generators and the image of the first supplies finitely many generators for all relations. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 12.17 (Hom is left exact proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $0 \rightarrow N' \rightarrow N \rightarrow N''$, the sequence $0 \rightarrow \text{Hom}(M, N') \rightarrow \text{Hom}(M, N) \rightarrow \text{Hom}(M, N'')$ is exact.

Solution. A map into N lands in N' exactly when its composite with $N \rightarrow N''$ is zero. Injectivity follows because $N' \rightarrow N$ is injective. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 12.18 (Localization of Hom for finitely presented modules proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If M is finitely presented, then $S^{-1}\text{Hom}_A(M, N) \cong \text{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N)$.

Solution. Choose a finite presentation of M , apply Hom to it, localize the resulting kernel description, and use exactness of localization together with the evident formula for finite free modules. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 12.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Hom and finite presentation control maps out of modules and the stability of algebraic data under limits and base change. Finitely presented modules are those described by finitely many generators and finitely many relations. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

12.14 Exercises with complete solutions

Exercise 12.20. State and apply the central principle behind Hom modules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For A -modules M, N , $\text{Hom}_A(M, N)$ is an A -module under pointwise operations. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 12.21. State and apply the central principle behind Contravariance and covariance. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Hom is contravariant in its first argument and covariant in its second. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 12.22. State and apply the central principle behind Left exactness. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\text{Hom}_A(M, -)$ preserves kernels but not generally surjections. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 12.23. State and apply the central principle behind Finite generation. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A module is finitely generated if a finite free module surjects onto it. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 12.24. State and apply the central principle behind Finite presentation. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A module is finitely presented if it is the cokernel of a map between finite free modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 12.25. State and apply the central principle behind Generators and relations. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A finite presentation packages finitely many generators together with finitely many module relations. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 12.26. State and apply the central principle behind Hom and filtered colimits. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Finite presentation is characterized by commuting of Hom out of the module with filtered colimits. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 12.27. State and apply the central principle behind Base change of presentations. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Finite presentations remain finite after scalar extension. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 13

Free and Projective Modules

Free modules have bases, while projective modules are direct summands of free modules. Projectivity is exactly the lifting property that makes Hom out of the module preserve surjections.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

13.1 Free modules

A free module is a direct sum of copies of the ring with a distinguished basis.

13.2 Rank

Over suitable rings, finite free modules have a well-defined rank; locally, projective modules acquire ranks prime by prime.

13.3 Projective modules

A module is projective if maps to quotients lift across surjections.

13.4 Direct summands

Projective modules are exactly direct summands of free modules.

13.5 Idempotent matrices

Finite projective modules arise as images of idempotent endomorphisms of finite free modules.

13.6 Splitting exact sequences

A surjection onto a projective module splits.

13.7 Hom exactness

P is projective iff $\text{Hom}(P, -)$ is exact.

13.8 Localization

Localization preserves projectivity, and finite projective modules are locally free.

13.9 Examples

Free modules, ideals generated by idempotents, and vector bundles over affine rings illustrate projectivity.

13.10 Core structural results

Theorem 13.1 (Projective summand criterion). *An A -module P is projective iff it is a direct summand of a free module.*

Proof. Choose a free module F surjecting onto P . If P is projective, the surjection splits, so $F \cong P \oplus K$. Conversely direct summands of free modules inherit the lifting property. $\square$

Theorem 13.2 (Splitting criterion). *A module P is projective iff every short exact sequence $0 \rightarrow K \rightarrow M \rightarrow P \rightarrow 0$ splits.*

Proof. Projectivity lifts the identity map of P through the surjection $M \rightarrow P$, producing a section. Conversely apply the splitting property to pullbacks of arbitrary surjections. $\square$

Theorem 13.3 (Hom exactness criterion). *P is projective iff $\text{Hom}_A(P, -)$ is an exact functor.*

Proof. Hom is always left exact. Exactness on surjections is precisely the lifting property defining projectivity. $\square$

13.11 Worked examples

Example 13.4 (Free modules diagnostic). Give a rigorous worked analysis of Free modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A free module is a direct sum of copies of the ring with a distinguished basis. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 13.5 (Rank diagnostic). Give a rigorous worked analysis of Rank. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Over suitable rings, finite free modules have a well-defined rank; locally, projective modules acquire ranks prime by prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 13.6 (Projective modules diagnostic). Give a rigorous worked analysis of Projective modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A module is projective if maps to quotients lift across surjections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 13.7 (Direct summands diagnostic). Give a rigorous worked analysis of Direct summands. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Projective modules are exactly direct summands of free modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

13.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 13.8 (Free modules diagnostic). Give a rigorous worked analysis of Free modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A free module is a direct sum of copies of the ring with a distinguished basis. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 13.9 (Rank diagnostic). Give a rigorous worked analysis of Rank. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over suitable rings, finite free modules have a well-defined rank; locally, projective modules acquire ranks prime by prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 13.10 (Projective modules diagnostic). Give a rigorous worked analysis of Projective modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is projective if maps to quotients lift across surjections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 13.11 (Direct summands diagnostic). Give a rigorous worked analysis of Direct summands. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Projective modules are exactly direct summands of free modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 13.12 (Idempotent matrices diagnostic). Give a rigorous worked analysis of Idempotent matrices. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite projective modules arise as images of idempotent endomorphisms of finite free modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 13.13 (Splitting exact sequences diagnostic). Give a rigorous worked analysis of Splitting exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A surjection onto a projective module splits. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 13.14 (Hom exactness diagnostic). Give a rigorous worked analysis of Hom exactness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. P is projective iff $\text{Hom}(P, -)$ is exact. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 13.15 (Localization diagnostic). Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization preserves projectivity, and finite projective modules are locally free. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 13.16 (Projective summand criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: An A -module P is projective iff it is a direct summand of a free module.

Solution. Choose a free module F surjecting onto P . If P is projective, the surjection splits, so $F \cong P \oplus K$. Conversely direct summands of free modules inherit the lifting property. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 13.17 (Splitting criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A module P is projective iff every short exact sequence $0 \rightarrow K \rightarrow M \rightarrow P \rightarrow 0$ splits.

Solution. Projectivity lifts the identity map of P through the surjection $M \rightarrow P$, producing a section. Conversely apply the splitting property to pullbacks of arbitrary surjections. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 13.18 (Hom exactness criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: P is projective iff $\text{Hom}_A(P, -)$ is an exact functor.

Solution. Hom is always left exact. Exactness on surjections is precisely the lifting property defining projectivity. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 13.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Free modules have bases, while projective modules are direct summands of free modules. Projectivity is exactly the lifting property that makes Hom out of the module preserve surjections. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

13.13 Exercises with complete solutions

Exercise 13.20. State and apply the central principle behind Free modules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A free module is a direct sum of copies of the ring with a distinguished basis. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 13.21. State and apply the central principle behind Rank. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Over suitable rings, finite free modules have a well-defined rank; locally, projective modules acquire ranks prime by prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 13.22. State and apply the central principle behind Projective modules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A module is projective if maps to quotients lift across surjections. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 13.23. State and apply the central principle behind Direct summands. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Projective modules are exactly direct summands of free modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 13.24. State and apply the central principle behind Idempotent matrices. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Finite projective modules arise as images of idempotent endomorphisms of finite free modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 13.25. State and apply the central principle behind Splitting exact sequences. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A surjection onto a projective module splits. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 13.26. State and apply the central principle behind Hom exactness. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. P is projective iff $\text{Hom}(P, -)$ is exact. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 13.27. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Localization preserves projectivity, and finite projective modules are locally free. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 14

Flat Modules

Flat modules are those whose tensor product functor preserves injections and hence all finite exact sequences. Flatness is the homological condition behind well-behaved base change.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

14.1 Definition of flatness

An A -module F is flat if $- \otimes_A F$ is exact.

14.2 Injection test

Because tensor is always right exact, flatness is equivalent to preserving injections.

14.3 Free and projective modules

Free modules are flat, and direct summands of flat modules are flat; hence projectives are flat.

14.4 Localization

Every localization $S^{-1}A$ is flat over A .

14.5 Filtered colimits

Filtered colimits of flat modules are flat.

14.6 Ideal criterion

Flatness can be tested by injectivity of $I \otimes F \rightarrow F$ for finitely generated ideals under standard hypotheses.

14.7 Tor preview

$\mathrm{Tor}_1^A(-, F)$ measures the failure of flatness.

14.8 Faithful flatness

A flat module is faithfully flat when tensoring also detects nonzero modules.

14.9 Base change

Flat scalar extension preserves short exact sequences and scheme-theoretic fiber behavior.

14.10 Examples and failures

Quotient modules are generally not flat; localization provides the basic positive example.

14.11 Core structural results

Theorem 14.1 (Free implies flat). *Every free A -module is flat.*

Proof. Tensoring with a direct sum of copies of A gives the corresponding direct sum of the original sequence. Direct sums preserve exactness in modules. $\square$

Theorem 14.2 (Projective implies flat). *Every projective module is flat.*

Proof. A projective module is a direct summand of a free module. Tensoring with the corresponding direct-sum decomposition shows the tensor functor for the projective summand is a direct summand of an exact functor and hence exact. $\square$

Theorem 14.3 (Localization is flat). *$S^{-1}A$ is flat as an A -module.*

Proof. Tensoring with $S^{-1}A$ is localization of modules, and localization is exact. $\square$

14.12 Worked examples

Example 14.4 (Definition of flatness diagnostic). Give a rigorous worked analysis of Definition of flatness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An A -module F is flat if $- \otimes_A F$ is exact. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 14.5 (Injection test diagnostic). Give a rigorous worked analysis of Injection test. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Because tensor is always right exact, flatness is equivalent to preserving injections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 14.6 (Free and projective modules diagnostic). Give a rigorous worked analysis of Free and projective modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Free modules are flat, and direct summands of flat modules are flat; hence projectives are flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 14.7 (Localization diagnostic). Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every localization $S^{-1}A$ is flat over A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

14.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 14.8 (Definition of flatness diagnostic). Give a rigorous worked analysis of Definition of flatness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An A -module F is flat if $- \otimes_A F$ is exact. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 14.9 (Injection test diagnostic). Give a rigorous worked analysis of Injection test. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Because tensor is always right exact, flatness is equivalent to preserving injections. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 14.10 (Free and projective modules diagnostic). Give a rigorous worked analysis of Free and projective modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Free modules are flat, and direct summands of flat modules are flat; hence projectives are flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 14.11 (Localization diagnostic). Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every localization $S^{-1}A$ is flat over A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 14.12 (Filtered colimits diagnostic). Give a rigorous worked analysis of Filtered colimits. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Filtered colimits of flat modules are flat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 14.13 (Ideal criterion diagnostic). Give a rigorous worked analysis of Ideal criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Flatness can be tested by injectivity of $I \otimes F \rightarrow F$ for finitely generated ideals under standard hypotheses. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 14.14 (Tor preview diagnostic). Give a rigorous worked analysis of Tor preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\text{Tor}_1^A(-, F)$ measures the failure of flatness. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 14.15 (Faithful flatness diagnostic). Give a rigorous worked analysis of Faithful flatness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A flat module is faithfully flat when tensoring also detects nonzero modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 14.16 (Free implies flat proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every free A -module is flat.

Solution. Tensoring with a direct sum of copies of A gives the corresponding direct sum of the original sequence. Direct sums preserve exactness in modules. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 14.17 (Projective implies flat proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every projective module is flat.

Solution. A projective module is a direct summand of a free module. Tensoring with the corresponding direct-sum decomposition shows the tensor functor for the projective summand is a direct summand of an exact functor and hence exact. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 14.18 (Localization is flat proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: $S^{-1}A$ is flat as an A -module.

Solution. Tensoring with $S^{-1}A$ is localization of modules, and localization is exact. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 14.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Flat modules are those whose tensor product functor preserves injections and hence all finite exact sequences. Flatness is the homological condition behind well-behaved base change. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

14.14 Exercises with complete solutions

Exercise 14.20. State and apply the central principle behind Definition of flatness. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An A -module F is flat if $- \otimes_A F$ is exact. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 14.21. State and apply the central principle behind Injection test. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Because tensor is always right exact, flatness is equivalent to preserving injections. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 14.22. State and apply the central principle behind Free and projective modules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Free modules are flat, and direct summands of flat modules are flat; hence projectives are flat. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 14.23. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Every localization $S^{-1}A$ is flat over A . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 14.24. State and apply the central principle behind Filtered colimits. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Filtered colimits of flat modules are flat. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 14.25. State and apply the central principle behind Ideal criterion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Flatness can be tested by injectivity of $I \otimes F \rightarrow F$ for finitely generated ideals under standard hypotheses. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 14.26. State and apply the central principle behind Tor preview. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\mathrm{Tor}_1^A(-, F)$ measures the failure of flatness. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 14.27. State and apply the central principle behind Faithful flatness. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A flat module is faithfully flat when tensoring also detects nonzero modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Part IV

Noetherian Algebra

Chapter 15

Noetherian Rings and Modules

Noetherian conditions impose finiteness on ideals and submodules. They are stable under quotients and finite-type polynomial extension and support induction arguments throughout algebraic geometry.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

15.1 ACC on submodules

A module is Noetherian when every ascending chain of submodules stabilizes.

15.2 Finite-generation criterion

A module is Noetherian iff every submodule is finitely generated.

15.3 Noetherian rings

A ring is Noetherian when it is Noetherian as a module over itself, equivalently every ideal is finitely generated.

15.4 Exact sequences

In a short exact sequence, the middle module is Noetherian iff both ends are Noetherian.

15.5 Finite modules

Finite modules over a Noetherian ring are Noetherian.

15.6 Quotients

Quotients of Noetherian rings and modules are Noetherian.

15.7 Hilbert basis theorem

If A is Noetherian, then $A[x]$ is Noetherian.

15.8 Finite-type algebras

Iterating Hilbert's theorem shows finitely generated A -algebras are Noetherian.

15.9 Noetherian induction

Properties can be proved by assuming all strictly larger subobjects satisfy them.

15.10 Examples

Fields, PID-like rings, polynomial rings over fields, and many coordinate rings are Noetherian.

15.11 Core structural results

Theorem 15.1 (Finite-generation criterion). *An A -module is Noetherian iff every submodule is finitely generated.*

Proof. If every submodule is finite, the union of an ascending chain is finitely generated and therefore contained in one stage. Conversely, if a submodule were not finitely generated, repeatedly adjoining new elements would produce a strictly increasing chain. $\square$

Theorem 15.2 (Exact-sequence criterion). *For $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$, M is Noetherian iff M' and M'' are Noetherian.*

Proof. Submodules of M' and images in M'' control any chain in M . For the converse, submodules and quotients of a Noetherian module are Noetherian. $\square$

Theorem 15.3 (Hilbert basis theorem). *If A is Noetherian, then $A[x]$ is Noetherian.*

Proof. For an ideal $I \subset A[x]$, leading coefficients form an ideal of A and stabilize under finite generation. Choose finitely many polynomials controlling high-degree leading terms and finitely many lower-degree remainders; together they generate I . $\square$

15.12 Worked examples

Example 15.4 (ACC on submodules diagnostic). Give a rigorous worked analysis of ACC on submodules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A module is Noetherian when every ascending chain of submodules stabilizes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 15.5 (Finite-generation criterion diagnostic). Give a rigorous worked analysis of Finite-generation criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A module is Noetherian iff every submodule is finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 15.6 (Noetherian rings diagnostic). Give a rigorous worked analysis of Noetherian rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A ring is Noetherian when it is Noetherian as a module over itself, equivalently every ideal is finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 15.7 (Exact sequences diagnostic). Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. In a short exact sequence, the middle module is Noetherian iff both ends are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

15.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 15.8 (ACC on submodules diagnostic). Give a rigorous worked analysis of ACC on submodules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is Noetherian when every ascending chain of submodules stabilizes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 15.9 (Finite-generation criterion diagnostic). Give a rigorous worked analysis of Finite-generation criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is Noetherian iff every submodule is finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 15.10 (Noetherian rings diagnostic). Give a rigorous worked analysis of Noetherian rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A ring is Noetherian when it is Noetherian as a module over itself, equivalently every ideal is finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 15.11 (Exact sequences diagnostic). Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. In a short exact sequence, the middle module is Noetherian iff both ends are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 15.12 (Finite modules diagnostic). Give a rigorous worked analysis of Finite modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite modules over a Noetherian ring are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 15.13 (Quotients diagnostic). Give a rigorous worked analysis of Quotients. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Quotients of Noetherian rings and modules are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 15.14 (Hilbert basis theorem diagnostic). Give a rigorous worked analysis of Hilbert basis theorem. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. If A is Noetherian, then $A[x]$ is Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 15.15 (Finite-type algebras diagnostic). Give a rigorous worked analysis of Finite-type algebras. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Iterating Hilbert's theorem shows finitely generated A -algebras are Noetherian. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 15.16 (Finite-generation criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: An A -module is Noetherian iff every submodule is finitely generated.

Solution. If every submodule is finite, the union of an ascending chain is finitely generated and therefore contained in one stage. Conversely, if a submodule were not finitely generated, repeatedly adjoining new elements would produce a strictly increasing chain. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 15.17 (Exact-sequence criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$, M is Noetherian iff M' and M'' are Noetherian.

Solution. Submodules of M' and images in M'' control any chain in M . For the converse, submodules and quotients of a Noetherian module are Noetherian. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 15.18 (Hilbert basis theorem proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If A is Noetherian, then $A[x]$ is Noetherian.

Solution. For an ideal $I \subset A[x]$, leading coefficients form an ideal of A and stabilize under finite generation. Choose finitely many polynomials controlling high-degree leading terms and finitely many lower-degree remainders; together they generate I . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 15.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Noetherian conditions impose finiteness on ideals and submodules. They are stable under quotients and finite-type polynomial extension and support induction arguments throughout algebraic geometry. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

15.14 Exercises with complete solutions

Exercise 15.20. State and apply the central principle behind ACC on submodules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A module is Noetherian when every ascending chain of submodules stabilizes. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 15.21. State and apply the central principle behind Finite-generation criterion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A module is Noetherian iff every submodule is finitely generated. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 15.22. State and apply the central principle behind Noetherian rings. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A ring is Noetherian when it is Noetherian as a module over itself, equivalently every ideal is finitely generated. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 15.23. State and apply the central principle behind Exact sequences. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. In a short exact sequence, the middle module is Noetherian iff both ends are Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 15.24. State and apply the central principle behind Finite modules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Finite modules over a Noetherian ring are Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 15.25. State and apply the central principle behind Quotients. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Quotients of Noetherian rings and modules are Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 15.26. State and apply the central principle behind Hilbert basis theorem. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. If A is Noetherian, then $A[x]$ is Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 15.27. State and apply the central principle behind Finite-type algebras. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Iterating Hilbert's theorem shows finitely generated A -algebras are Noetherian. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 16

Support

The support of a module records the primes where the module remains nonzero after localization. It turns module-theoretic information into a subset of the prime spectrum and behaves well under exact sequences.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

16.1 Definition

$$\operatorname{Supp} M = \{\mathfrak{p} : M_{\mathfrak{p}} \neq 0\}.$$

16.2 Annihilator

The annihilator $\operatorname{Ann}(M)$ consists of scalars killing every element of M .

16.3 Finite-module support

For finitely generated M , $\operatorname{Supp} M = V(\operatorname{Ann} M)$.

16.4 Cyclic modules

$$\operatorname{Supp}(A/I) = V(I).$$

16.5 Exact sequences

Support of the middle term in a short exact sequence is the union of supports of the ends.

16.6 Direct sums

Support of a direct sum is the union of the supports.

16.7 Tensor products

Support of a tensor product is contained in the intersection of supports and equals it under common finite-generation hypotheses over local rings.

16.8 Localization

Support localizes by restricting to primes contained in the chosen localization domain.

16.9 Vanishing detection

A module is zero iff all localizations at maximal ideals are zero.

16.10 Geometric role

Support marks the locus where a sheaf-like algebraic object is present.

16.11 Core structural results

Theorem 16.1 (Support of a finite module). *If M is finitely generated, then $\text{Supp } M = V(\text{Ann } M)$.*

Proof. If an annihilator element becomes a unit at $\mathfrak{p}$, the localized module vanishes. Conversely, if $M_{\mathfrak{p}} = 0$, finitely many generators are each killed by denominators outside $\mathfrak{p}$; their product lies in $\text{Ann } M$ but outside $\mathfrak{p}$. $\square$

Theorem 16.2 (Support in a short exact sequence). *For $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$, $\text{Supp } M = \text{Supp } M' \cup \text{Supp } M''$.*

Proof. Localize the sequence. Exactness of localization shows the middle localization is zero exactly when both end localizations are zero. $\square$

Theorem 16.3 (Maximal-local detection). *An A -module M is zero iff $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$.*

Proof. If $m \neq 0$, its annihilator is proper and lies in some maximal ideal $\mathfrak{m}$. Then $m/1$ cannot vanish in $M_{\mathfrak{m}}$, otherwise an element outside $\mathfrak{m}$ would annihilate m . $\square$

16.12 Worked examples

Example 16.4 (Definition diagnostic). Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\text{Supp } M = \{\mathfrak{p} : M_{\mathfrak{p}} \neq 0\}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 16.5 (Annihilator diagnostic). Give a rigorous worked analysis of Annihilator. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The annihilator $\text{Ann}(M)$ consists of scalars killing every element of M . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 16.6 (Finite-module support diagnostic). Give a rigorous worked analysis of Finite-module support. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For finitely generated M , $\text{Supp } M = V(\text{Ann } M)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 16.7 (Cyclic modules diagnostic). Give a rigorous worked analysis of Cyclic modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\text{Supp}(A/I) = V(I)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

16.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 16.8 (Definition diagnostic). Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\text{Supp } M = \{\mathfrak{p} : M_{\mathfrak{p}} \neq 0\}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 16.9 (Annihilator diagnostic). Give a rigorous worked analysis of Annihilator. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The annihilator $\text{Ann}(M)$ consists of scalars killing every element of M . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 16.10 (Finite-module support diagnostic). Give a rigorous worked analysis of Finite-module support. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For finitely generated M , $\text{Supp } M = V(\text{Ann } M)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 16.11 (Cyclic modules diagnostic). Give a rigorous worked analysis of Cyclic modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\text{Supp}(A/I) = V(I)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 16.12 (Exact sequences diagnostic). Give a rigorous worked analysis of Exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Support of the middle term in a short exact sequence is the union of supports of the ends. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 16.13 (Direct sums diagnostic). Give a rigorous worked analysis of Direct sums. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Support of a direct sum is the union of the supports. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 16.14 (Tensor products diagnostic). Give a rigorous worked analysis of Tensor products. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Support of a tensor product is contained in the intersection of supports and equals it under common finite-generation hypotheses over local rings. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 16.15 (Localization diagnostic). Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Support localizes by restricting to primes contained in the chosen localization domain. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 16.16 (Support of a finite module proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If M is finitely generated, then $\text{Supp } M = V(\text{Ann } M)$.

Solution. If an annihilator element becomes a unit at $\mathfrak{p}$, the localized module vanishes. Conversely, if $M_{\mathfrak{p}} = 0$, finitely many generators are each killed by denominators outside $\mathfrak{p}$; their product lies in $\text{Ann } M$ but outside $\mathfrak{p}$. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 16.17 (Support in a short exact sequence proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$, $\text{Supp } M = \text{Supp } M' \cup \text{Supp } M''$.

Solution. Localize the sequence. Exactness of localization shows the middle localization is zero exactly when both end localizations are zero. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 16.18 (Maximal-local detection proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: An A -module M is zero iff $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$.

Solution. If $m \neq 0$, its annihilator is proper and lies in some maximal ideal $\mathfrak{m}$. Then $m/1$ cannot vanish in $M_{\mathfrak{m}}$, otherwise an element outside $\mathfrak{m}$ would annihilate m . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 16.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. The support of a module records the primes where the module remains nonzero after localization. It turns module-theoretic information into a subset of the prime spectrum and behaves well under exact sequences. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

16.14 Exercises with complete solutions

Exercise 16.20. State and apply the central principle behind Definition. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\text{Supp } M = \{\mathfrak{p} : M_{\mathfrak{p}} \neq 0\}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 16.21. State and apply the central principle behind Annihilator. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The annihilator $\text{Ann}(M)$ consists of scalars killing every element of M . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 16.22. State and apply the central principle behind Finite-module support. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For finitely generated M , $\text{Supp } M = V(\text{Ann } M)$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 16.23. State and apply the central principle behind Cyclic modules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\text{Supp}(A/I) = V(I)$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 16.24. State and apply the central principle behind Exact sequences. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Support of the middle term in a short exact sequence is the union of supports of the ends. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 16.25. State and apply the central principle behind Direct sums. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Support of a direct sum is the union of the supports. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 16.26. State and apply the central principle behind Tensor products. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Support of a tensor product is contained in the intersection of supports and equals it under common finite-generation hypotheses over local rings. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 16.27. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Support localizes by restricting to primes contained in the chosen localization domain. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 17

Associated Primes

Associated primes identify prime annihilators of individual module elements. They refine support by detecting the primes that occur as embedded or minimal algebraic components.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

17.1 Definition

A prime $\mathfrak{p}$ is associated to M if $\mathfrak{p} = \text{Ann}(m)$ for some $m \in M$.

17.2 Cyclic submodules

Associated primes arise exactly from embedded copies of $A/\mathfrak{p}$ inside M .

17.3 Existence in Noetherian modules

Every nonzero finitely generated module over a Noetherian ring has an associated prime.

17.4 Containment in support

Every associated prime belongs to the support.

17.5 Minimal support primes

Minimal primes in the support of a finite module are associated.

17.6 Exact sequence behavior

Associated primes of ends and middle terms satisfy useful inclusions.

17.7 Localization

Associated primes localize by keeping precisely those contained in the localization prime.

17.8 Zero divisors

For finite modules over Noetherian rings, the zero divisors on M are the union of associated primes.

17.9 Primary decomposition preview

Associated primes index components that later appear in primary decompositions.

17.10 Core structural results

Theorem 17.1 (Associated prime existence). *If $M \neq 0$ is finitely generated over a Noetherian ring, then M has an associated prime.*

Proof. Choose an annihilator maximal among annihilators of nonzero elements. If ab lies in it and b does not, the annihilator of bm contains the chosen annihilator and also a ; maximality forces a into the original annihilator, proving primality. $\square$

Theorem 17.2 (Associated implies support). *If $\mathfrak{p} \in \text{Ass } M$, then $\mathfrak{p} \in \text{Supp } M$.*

Proof. An embedding $A/\mathfrak{p} \hookrightarrow M$ localizes to a nonzero embedding at $\mathfrak{p}$, since $(A/\mathfrak{p})_{\mathfrak{p}}$ is nonzero. $\square$

Theorem 17.3 (Minimal support primes are associated). *For a finitely generated module over a Noetherian ring, every prime minimal in $\text{Supp } M$ lies in $\text{Ass } M$.*

Proof. Localize at the minimal prime. The localized support has only the maximal prime, so an associated prime of the nonzero localized module must be that maximal prime; contract back. $\square$

17.11 Worked examples

Example 17.4 (Definition diagnostic). Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A prime $\mathfrak{p}$ is associated to M if $\mathfrak{p} = \text{Ann}(m)$ for some $m \in M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 17.5 (Cyclic submodules diagnostic). Give a rigorous worked analysis of Cyclic submodules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Associated primes arise exactly from embedded copies of $A/\mathfrak{p}$ inside M . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 17.6 (Existence in Noetherian modules diagnostic). Give a rigorous worked analysis of Existence in Noetherian modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every nonzero finitely generated module over a Noetherian ring has an associated prime. The decisive step is to use the

universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 17.7 (Containment in support diagnostic). Give a rigorous worked analysis of Containment in support. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every associated prime belongs to the support. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

17.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 17.8 (Definition diagnostic). Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A prime $\mathfrak{p}$ is associated to M if $\mathfrak{p} = \text{Ann}(m)$ for some $m \in M$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 17.9 (Cyclic submodules diagnostic). Give a rigorous worked analysis of Cyclic submodules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Associated primes arise exactly from embedded copies of $A/\mathfrak{p}$ inside M . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 17.10 (Existence in Noetherian modules diagnostic). Give a rigorous worked analysis of Existence in Noetherian modules. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every nonzero finitely generated module over a Noetherian ring has an associated prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 17.11 (Containment in support diagnostic). Give a rigorous worked analysis of Containment in support. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every associated prime belongs to the support. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 17.12 (Minimal support primes diagnostic). Give a rigorous worked analysis of Minimal support primes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Minimal primes in the support of a finite module are associated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 17.13 (Exact sequence behavior diagnostic). Give a rigorous worked analysis of Exact sequence behavior. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Associated primes of ends and middle terms satisfy useful inclusions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 17.14 (Localization diagnostic). Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Associated primes localize by keeping precisely those contained in the localization prime. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 17.15 (Zero divisors diagnostic). Give a rigorous worked analysis of Zero divisors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For finite modules over Noetherian rings, the zero divisors on M are the union of associated primes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 17.16 (Associated prime existence proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $M \neq 0$ is finitely generated over a Noetherian ring, then M has an associated prime.

Solution. Choose an annihilator maximal among annihilators of nonzero elements. If ab lies in it and b does not, the annihilator of bm contains the chosen annihilator and also a ; maximality forces a into the original annihilator, proving primality. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 17.17 (Associated implies support proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $\mathfrak{p} \in \text{Ass } M$, then $\mathfrak{p} \in \text{Supp } M$.

Solution. An embedding $A/\mathfrak{p} \hookrightarrow M$ localizes to a nonzero embedding at $\mathfrak{p}$, since $(A/\mathfrak{p})_{\mathfrak{p}}$ is nonzero. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 17.18 (Minimal support primes are associated proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For a finitely generated module over a Noetherian ring, every prime minimal in $\text{Supp } M$ lies in $\text{Ass } M$.

Solution. Localize at the minimal prime. The localized support has only the maximal prime, so an associated prime of the nonzero localized module must be that maximal prime; contract back. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 17.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Associated primes identify prime annihilators of individual module elements. They refine support by detecting the primes that occur as embedded or minimal algebraic components. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

17.13 Exercises with complete solutions

Exercise 17.20. State and apply the central principle behind Definition. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A prime $\mathfrak{p}$ is associated to M if $\mathfrak{p} = \text{Ann}(m)$ for some $m \in M$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 17.21. State and apply the central principle behind Cyclic submodules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Associated primes arise exactly from embedded copies of $A/\mathfrak{p}$ inside M . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 17.22. State and apply the central principle behind Existence in Noetherian modules. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Every nonzero finitely generated module over a Noetherian ring has an associated prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 17.23. State and apply the central principle behind Containment in support. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Every associated prime belongs to the support. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 17.24. State and apply the central principle behind Minimal support primes. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Minimal primes in the support of a finite module are associated. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 17.25. State and apply the central principle behind Exact sequence behavior. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Associated primes of ends and middle terms satisfy useful inclusions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 17.26. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Associated primes localize by keeping precisely those contained in the localization prime. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 17.27. State and apply the central principle behind Zero divisors. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For finite modules over Noetherian rings, the zero divisors on M are the union of associated primes. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 18

Completion and I-Adic Topology

I-adic topology and completion encode infinite approximation by powers of an ideal. Completion turns compatible residue classes modulo I^n into actual elements and is fundamental for local algebra and formal geometry.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

18.1 I-adic filtration

The powers $M \supset IM \supset I^2M \supset \cdots$ define progressively finer congruences.

18.2 I-adic topology

Cosets of I^nM form neighborhoods of module elements.

18.3 Separatedness

The topology is separated when $\bigcap_n I^nM = 0$.

18.4 Completion

$\widehat{M} = \varprojlim_n M/I^nM$ consists of compatible systems of residues.

18.5 Ring completion

$\widehat{A}$ is a ring and $\widehat{M}$ naturally a module over it.

18.6 Universal approximation

A completion element determines all finite-order approximations simultaneously.

18.7 Examples

p -adic integers and formal power-series rings arise as completions.

18.8 Krull intersection preview

Under Noetherian local hypotheses, intersections of powers often vanish.

18.9 Exactness issues

Completion is not always exact, but for finite modules over Noetherian rings it behaves much better.

18.10 Comparison with tensor

For finitely generated modules in the Noetherian setting, completion often agrees with tensoring by $\hat{A}$.

18.11 Core structural results

Theorem 18.1 (Inverse-limit description). *The I -adic completion of M is the inverse limit $\varprojlim M/I^n M$.*

Proof. A Cauchy-compatible system is exactly a sequence of residue classes whose class modulo $I^n M$ agrees with every later approximation. This is the defining inverse-limit condition. $\square$

Theorem 18.2 (Completion map kernel). *The kernel of the natural map $M \rightarrow \hat{M}$ is $\bigcap_{n \geq 1} I^n M$.*

Proof. An element maps to zero in the inverse limit exactly when it is zero in every quotient $M/I^n M$, equivalently when it lies in every $I^n M$. $\square$

Theorem 18.3 (Power-series completion). *The (x) -adic completion of $k[x]$ is canonically $k[[x]]$.*

Proof. A compatible sequence modulo x^n uniquely specifies all coefficients of a formal power series, and truncating a power series gives a compatible residue system. $\square$

18.12 Worked examples

Example 18.4 (I -adic filtration diagnostic). Give a rigorous worked analysis of I -adic filtration. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The powers $M \supset IM \supset I^2 M \supset \cdots$ define progressively finer congruences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 18.5 (I-adic topology diagnostic). Give a rigorous worked analysis of I-adic topology. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Cosets of $I^n M$ form neighborhoods of module elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 18.6 (Separatedness diagnostic). Give a rigorous worked analysis of Separatedness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The topology is separated when $\bigcap_n I^n M = 0$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 18.7 (Completion diagnostic). Give a rigorous worked analysis of Completion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\widehat{M} = \varprojlim_n M/I^n M$ consists of compatible systems of residues. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

18.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 18.8 (I-adic filtration diagnostic). Give a rigorous worked analysis of I-adic filtration. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The powers $M \supset IM \supset I^2 M \supset \cdots$ define progressively finer congruences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.9 (I-adic topology diagnostic). Give a rigorous worked analysis of I-adic topology. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Cosets of $I^n M$ form neighborhoods of module elements. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.10 (Separatedness diagnostic). Give a rigorous worked analysis of Separatedness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The topology is separated when $\bigcap_n I^n M = 0$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.11 (Completion diagnostic). Give a rigorous worked analysis of Completion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\widehat{M} = \varprojlim_n M/I^n M$ consists of compatible systems of residues. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.12 (Ring completion diagnostic). Give a rigorous worked analysis of Ring completion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\widehat{A}$ is a ring and $\widehat{M}$ naturally a module over it. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.13 (Universal approximation diagnostic). Give a rigorous worked analysis of Universal approximation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A completion element determines all finite-order approximations simultaneously. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.14 (Examples diagnostic). Give a rigorous worked analysis of Examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. p -adic integers and formal power-series rings arise as completions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.15 (Krull intersection preview diagnostic). Give a rigorous worked analysis of Krull intersection preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Under Noetherian local hypotheses, intersections of powers often vanish. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 18.16 (Inverse-limit description proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: The I -adic completion of M is the inverse limit $\varprojlim M/I^n M$.

Solution. A Cauchy-compatible system is exactly a sequence of residue classes whose class modulo $I^n M$ agrees with every later approximation. This is the defining inverse-limit condition. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 18.17 (Completion map kernel proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: The kernel of the natural map $M \rightarrow \widehat{M}$ is $\bigcap_{n \geq 1} I^n M$.

Solution. An element maps to zero in the inverse limit exactly when it is zero in every quotient $M/I^n M$, equivalently when it lies in every $I^n M$. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 18.18 (Power-series completion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: The (x) -adic completion of $k[x]$ is canonically $k[[x]]$.

Solution. A compatible sequence modulo x^n uniquely specifies all coefficients of a formal power series, and truncating a power series gives a compatible residue system. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 18.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. I-adic topology and completion encode infinite approximation by powers of an ideal. Completion turns compatible residue classes modulo I^n into actual elements and is fundamental for local algebra and formal geometry. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

18.14 Exercises with complete solutions

Exercise 18.20. State and apply the central principle behind I-adic filtration. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The powers $M \supset IM \supset I^2M \supset \cdots$ define progressively finer congruences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 18.21. State and apply the central principle behind I-adic topology. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Cosets of $I^n M$ form neighborhoods of module elements. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 18.22. State and apply the central principle behind Separatedness. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The topology is separated when $\bigcap_n I^n M = 0$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 18.23. State and apply the central principle behind Completion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\widehat{M} = \varprojlim_n M/I^n M$ consists of compatible systems of residues. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 18.24. State and apply the central principle behind Ring completion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\widehat{A}$ is a ring and $\widehat{M}$ naturally a module over it. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 18.25. State and apply the central principle behind Universal approximation. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A completion element determines all finite-order approximations simultaneously. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 18.26. State and apply the central principle behind Examples. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. p -adic integers and formal power-series rings arise as completions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 18.27. State and apply the central principle behind Krull intersection preview. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Under Noetherian local hypotheses, intersections of powers often vanish. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Part V

Integral and Valuation Theory

Chapter 19

Integral Dependence

Integral dependence generalizes algebraicity from field extensions to ring extensions. An element is integral when it satisfies a monic polynomial, and integrality is stable under finite algebraic constructions.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

19.1 Integral elements

An element b of an A -algebra B is integral over A if it satisfies a monic polynomial with coefficients in A .

19.2 Integral extensions

An extension $A \subset B$ is integral when every element of B is integral over A .

19.3 Finite module criterion

An element is integral iff the subalgebra it generates is finite as an A -module.

19.4 Finite algebras

Every finite A -algebra is integral over A .

19.5 Transitivity

Integrality is transitive through towers of ring extensions.

19.6 Integral elements form a ring

The set of elements of B integral over A is a subring.

19.7 Determinant trick

Finite-generation plus stability under multiplication produces monic equations by a determinant argument.

19.8 Prime behavior preview

Integral extensions satisfy lying-over and contraction properties for prime ideals.

19.9 Examples

Algebraic integers, roots of monic equations, and normalization rings illustrate integral dependence.

19.10 Core structural results

Theorem 19.1 (Finite module criterion). *For $b \in B$, b is integral over A iff $A[b]$ is finitely generated as an A -module.*

Proof. A monic equation expresses every high power of b in terms of lower powers. Conversely, if finitely many powers generate an A -module stable under multiplication by b , the determinant trick gives a monic polynomial annihilating b . $\square$

Theorem 19.2 (Finite algebra implies integral). *If B is a finite A -module and an A -algebra, then every $b \in B$ is integral over A .*

Proof. Multiplication by b is an A -linear endomorphism of the finite module B . Apply the determinant trick or Cayley–Hamilton to obtain a monic polynomial with coefficients in A annihilating b . $\square$

Theorem 19.3 (Transitivity of integrality). *If B is integral over A and C integral over B , then C is integral over A .*

Proof. For $c \in C$, its monic equation uses finitely many coefficients from B . Those coefficients generate a finite A -algebra because they are integral; adjoining c is finite over that finite algebra, hence finite over A , so c is integral over A . $\square$

19.11 Worked examples

Example 19.4 (Integral elements diagnostic). Give a rigorous worked analysis of Integral elements. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An element b of an A -algebra B is integral over A if it satisfies a monic polynomial with coefficients in A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 19.5 (Integral extensions diagnostic). Give a rigorous worked analysis of Integral extensions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An extension $A \subset B$ is integral when every element of B is integral over A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 19.6 (Finite module criterion diagnostic). Give a rigorous worked analysis of Finite module criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. An element is integral iff the subalgebra it generates is finite as an A -module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 19.7 (Finite algebras diagnostic). Give a rigorous worked analysis of Finite algebras. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Every finite A -algebra is integral over A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

19.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 19.8 (Integral elements diagnostic). Give a rigorous worked analysis of Integral elements. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element b of an A -algebra B is integral over A if it satisfies a monic polynomial with coefficients in A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 19.9 (Integral extensions diagnostic). Give a rigorous worked analysis of Integral extensions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An extension $A \subset B$ is integral when every element of B is integral over A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 19.10 (Finite module criterion diagnostic). Give a rigorous worked analysis of Finite module criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. An element is integral iff the subalgebra it generates is finite as an A -module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 19.11 (Finite algebras diagnostic). Give a rigorous worked analysis of Finite algebras. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every finite A -algebra is integral over A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 19.12 (Transitivity diagnostic). Give a rigorous worked analysis of Transitivity. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Integrality is transitive through towers of ring extensions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 19.13 (Integral elements form a ring diagnostic). Give a rigorous worked analysis of Integral elements form a ring. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The set of elements of B integral over A is a subring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 19.14 (Determinant trick diagnostic). Give a rigorous worked analysis of Determinant trick. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite-generation plus stability under multiplication produces monic equations by a determinant argument. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 19.15 (Prime behavior preview diagnostic). Give a rigorous worked analysis of Prime behavior preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Integral extensions satisfy lying-over and contraction properties for prime ideals. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 19.16 (Finite module criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For $b \in B$, b is integral over A iff $A[b]$ is finitely generated as an A -module.

Solution. A monic equation expresses every high power of b in terms of lower powers. Conversely, if finitely many powers generate an A -module stable under multiplication by b , the determinant trick gives a monic polynomial annihilating b . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 19.17 (Finite algebra implies integral proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If B is a finite A -module and an A -algebra, then every $b \in B$ is integral over A .

Solution. Multiplication by b is an A -linear endomorphism of the finite module B . Apply the determinant trick or Cayley–Hamilton to obtain a monic polynomial with coefficients in A annihilating b . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 19.18 (Transitivity of integrality proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If B is integral over A and C integral over B , then C is integral over A .

Solution. For $c \in C$, its monic equation uses finitely many coefficients from B . Those coefficients generate a finite A -algebra because they are integral; adjoining c is finite over that finite algebra, hence finite over A , so c is integral over A . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 19.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Integral dependence generalizes algebraicity from field extensions to ring extensions. An element is integral when it satisfies a monic polynomial, and integrality is stable under finite algebraic constructions. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

19.13 Exercises with complete solutions

Exercise 19.20. State and apply the central principle behind Integral elements. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An element b of an A -algebra B is integral over A if it satisfies a monic polynomial with coefficients in A . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 19.21. State and apply the central principle behind Integral extensions. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An extension $A \subset B$ is integral when every element of B is integral over A . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 19.22. State and apply the central principle behind Finite module criterion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. An element is integral iff the subalgebra it generates is finite as an A -module. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 19.23. State and apply the central principle behind Finite algebras. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Every finite A -algebra is integral over A . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 19.24. State and apply the central principle behind Transitivity. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Integrality is transitive through towers of ring extensions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 19.25. State and apply the central principle behind Integral elements form a ring. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The set of elements of B integral over A is a subring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 19.26. State and apply the central principle behind Determinant trick. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Finite-generation plus stability under multiplication produces monic equations by a determinant argument. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 19.27. State and apply the central principle behind Prime behavior preview. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Integral extensions satisfy lying-over and contraction properties for prime ideals. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 20

Integral Closure and Normalization

Integral closure collects all elements of a fraction field integral over the base ring. Normalization repairs non-normal domains by adjoining these integral elements.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

20.1 Integrally closed domains

A domain is integrally closed when every element of its fraction field integral over the ring already lies in the ring.

20.2 Integral closure

The integral closure of A in an extension field consists of all elements integral over A .

20.3 Normalization

For a domain, normalization is its integral closure in its fraction field.

20.4 UFDs and normality

Unique factorization domains are integrally closed.

20.5 Localization

Integral closure behaves well under localization.

20.6 Finite normalization

For many finitely generated algebras over fields, normalization is finite, though this requires hypotheses.

20.7 Singular examples

Cusp-like rings such as $k[t^2, t^3]$ normalize to $k[t]$.

20.8 Prime spectra

Normalization induces an integral map on spectra and separates certain singular algebraic behaviors.

20.9 Conductor

The conductor measures the largest ideal shared by a ring and its normalization.

20.10 Core structural results

Theorem 20.1 (UFDs are integrally closed). *Every unique factorization domain is integrally closed in its fraction field.*

Proof. Write an integral element as a/b in lowest terms. A monic equation implies a^n is divisible by b ; coprimality and unique factorization force b to be a unit. $\square$

Theorem 20.2 (Localization of integral closure). *If B is the integral closure of a domain A in a field extension and $S \subset A$ is multiplicative, then $S^{-1}B$ is the integral closure of $S^{-1}A$ in the same localized field.*

Proof. Localization preserves monic equations, giving one inclusion. For the reverse, clear finitely many denominators in a monic equation to show a suitable multiple of the element is integral over A , hence lies in B . $\square$

Theorem 20.3 (Normalization of a cusp ring). *The normalization of $k[t^2, t^3]$ in $k(t)$ is $k[t]$.*

Proof. The element t is integral because it satisfies $X^2 - t^2 = 0$ with coefficient t^2 in the subring. Since $k[t]$ is a PID and therefore integrally closed, no larger integral subring inside $k(t)$ is needed. $\square$

20.11 Worked examples

Example 20.4 (Integrally closed domains diagnostic). Give a rigorous worked analysis of Integrally closed domains. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A domain is integrally closed when every element of its fraction field integral over the ring already lies in the ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 20.5 (Integral closure diagnostic). Give a rigorous worked analysis of Integral closure. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The integral closure of A in an extension field consists of all elements integral over A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 20.6 (Normalization diagnostic). Give a rigorous worked analysis of Normalization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. For a domain, normalization is its integral closure in its fraction field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 20.7 (UFDs and normality diagnostic). Give a rigorous worked analysis of UFDs and normality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Unique factorization domains are integrally closed. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

20.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 20.8 (Integrally closed domains diagnostic). Give a rigorous worked analysis of Integrally closed domains. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A domain is integrally closed when every element of its fraction field integral over the ring already lies in the ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 20.9 (Integral closure diagnostic). Give a rigorous worked analysis of Integral closure. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The integral closure of A in an extension field consists of all elements integral over A . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 20.10 (Normalization diagnostic). Give a rigorous worked analysis of Normalization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For a domain, normalization is its integral closure in its fraction field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 20.11 (UFDs and normality diagnostic). Give a rigorous worked analysis of UFDs and normality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Unique factorization domains are integrally closed. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 20.12 (Localization diagnostic). Give a rigorous worked analysis of Localization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Integral closure behaves well under localization. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 20.13 (Finite normalization diagnostic). Give a rigorous worked analysis of Finite normalization. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For many finitely generated algebras over fields, normalization is finite, though this requires hypotheses. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 20.14 (Singular examples diagnostic). Give a rigorous worked analysis of Singular examples. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Cusp-like rings such as $k[t^2, t^3]$ normalize to $k[t]$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 20.15 (Prime spectra diagnostic). Give a rigorous worked analysis of Prime spectra. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Normalization induces an integral map on spectra and separates certain singular algebraic behaviors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 20.16 (UFDs are integrally closed proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every unique factorization domain is integrally closed in its fraction field.

Solution. Write an integral element as a/b in lowest terms. A monic equation implies a^n is divisible by b ; coprimality and unique factorization force b to be a unit. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 20.17 (Localization of integral closure proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If B is the integral closure of a domain A in a field extension and $S \subset A$ is multiplicative, then $S^{-1}B$ is the integral closure of $S^{-1}A$ in the same localized field.

Solution. Localization preserves monic equations, giving one inclusion. For the reverse, clear finitely many denominators in a monic equation to show a suitable multiple of the element is integral over A , hence lies in B . This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 20.18 (Normalization of a cusp ring proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: The normalization of $k[t^2, t^3]$ in $k(t)$ is $k[t]$.

Solution. The element t is integral because it satisfies $X^2 - t^2 = 0$ with coefficient t^2 in the subring. Since $k[t]$ is a PID and therefore integrally closed, no larger integral subring inside $k(t)$ is needed. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 20.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Integral closure collects all elements of a fraction field integral over the base ring. Normalization repairs non-normal domains by adjoining these integral elements. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

20.13 Exercises with complete solutions

Exercise 20.20. State and apply the central principle behind Integrally closed domains. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A domain is integrally closed when every element of its fraction field integral over the ring already lies in the ring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 20.21. State and apply the central principle behind Integral closure. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The integral closure of A in an extension field consists of all elements integral over A . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 20.22. State and apply the central principle behind Normalization. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For a domain, normalization is its integral closure in its fraction field. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 20.23. State and apply the central principle behind UFDs and normality. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Unique factorization domains are integrally closed. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 20.24. State and apply the central principle behind Localization. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Integral closure behaves well under localization. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 20.25. State and apply the central principle behind Finite normalization. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For many finitely generated algebras over fields, normalization is finite, though this requires hypotheses. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 20.26. State and apply the central principle behind Singular examples. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Cusp-like rings such as $k[t^2, t^3]$ normalize to $k[t]$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 20.27. State and apply the central principle behind Prime spectra. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Normalization induces an integral map on spectra and separates certain singular algebraic behaviors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 21

Valuation Rings

Valuation rings impose a total divisibility order on a field. They are integrally closed local domains and provide a powerful way to test integrality and dominate local rings.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

21.1 Valuations

A valuation assigns ordered-group values compatible with products and satisfying a triangle inequality for sums.

21.2 Valuation rings

A subring V of a field K is a valuation ring when for every nonzero $x \in K$, either x or x^{-1} lies in V .

21.3 Locality

The nonunits of a valuation ring form its unique maximal ideal.

21.4 Divisibility order

Principal ideals are totally ordered by inclusion in a valuation ring.

21.5 Discrete valuation rings

A DVR corresponds to a discrete rank-one valuation and has a principal maximal ideal.

21.6 Integrally closed property

Every valuation ring is integrally closed in its fraction field.

21.7 Dominating local rings

Valuation rings can contain a local domain while having maximal ideal contracting to the local maximal ideal.

21.8 Prime ideals

Prime ideals of a valuation ring are linearly ordered.

21.9 Integral closure criterion

Integral closure of a domain can be recovered as an intersection of valuation rings containing it.

21.10 Core structural results

Theorem 21.1 (Valuation rings are local). *If V is a valuation ring of a field, its nonunits form an ideal and hence the unique maximal ideal.*

Proof. For nonunits a, b , compare a/b or b/a when both are nonzero; one divides the other, so their sum is a multiple of one nonunit. Multiplication by arbitrary ring elements preserves nonunits. $\square$

Theorem 21.2 (Principal ideals are totally ordered). *For nonzero $a, b \in V$, either $(a) \subseteq (b)$ or $(b) \subseteq (a)$.*

Proof. Apply the valuation-ring condition to a/b : if $a/b \in V$, then $a \in (b)$; otherwise $b/a \in V$, so $b \in (a)$. $\square$

Theorem 21.3 (Valuation rings are integrally closed). *A valuation ring V is integrally closed in its fraction field.*

Proof. If x is integral over V but $x \notin V$, then $x^{-1} \in V$ and is a nonunit. Divide a monic equation for x by the highest power of x to express 1 as an element of the maximal ideal generated by powers of x^{-1} , contradiction. $\square$

21.11 Worked examples

Example 21.4 (Valuations diagnostic). Give a rigorous worked analysis of Valuations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A valuation assigns ordered-group values compatible with products and satisfying a triangle inequality for sums. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 21.5 (Valuation rings diagnostic). Give a rigorous worked analysis of Valuation rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A subring V of a field K is a valuation ring when for every nonzero $x \in K$, either x or x^{-1} lies in V . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 21.6 (Locality diagnostic). Give a rigorous worked analysis of Locality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The nonunits of a valuation ring form its unique maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 21.7 (Divisibility order diagnostic). Give a rigorous worked analysis of Divisibility order. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Principal ideals are totally ordered by inclusion in a valuation ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

21.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 21.8 (Valuations diagnostic). Give a rigorous worked analysis of Valuations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A valuation assigns ordered-group values compatible with products and satisfying a triangle inequality for sums. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 21.9 (Valuation rings diagnostic). Give a rigorous worked analysis of Valuation rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A subring V of a field K is a valuation ring when for every nonzero $x \in K$, either x or x^{-1} lies in V . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 21.10 (Locality diagnostic). Give a rigorous worked analysis of Locality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The nonunits of a valuation ring form its unique maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 21.11 (Divisibility order diagnostic). Give a rigorous worked analysis of Divisibility order. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Principal ideals are totally ordered by inclusion in a valuation ring. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 21.12 (Discrete valuation rings diagnostic). Give a rigorous worked analysis of Discrete valuation rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A DVR corresponds to a discrete rank-one valuation and has a principal maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 21.13 (Integrally closed property diagnostic). Give a rigorous worked analysis of Integrally closed property. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every valuation ring is integrally closed in its fraction field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 21.14 (Dominating local rings diagnostic). Give a rigorous worked analysis of Dominating local rings. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Valuation rings can contain a local domain while having maximal ideal contracting to the local maximal ideal. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 21.15 (Prime ideals diagnostic). Give a rigorous worked analysis of Prime ideals. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Prime ideals of a valuation ring are linearly ordered. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 21.16 (Valuation rings are local proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If V is a valuation ring of a field, its nonunits form an ideal and hence the unique maximal ideal.

Solution. For nonunits a, b , compare a/b or b/a when both are nonzero; one divides the other, so their sum is a multiple of one nonunit. Multiplication by arbitrary ring elements preserves nonunits. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 21.17 (Principal ideals are totally ordered proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For nonzero $a, b \in V$, either $(a) \subseteq (b)$ or $(b) \subseteq (a)$.

Solution. Apply the valuation-ring condition to a/b : if $a/b \in V$, then $a \in (b)$; otherwise $b/a \in V$, so $b \in (a)$. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 21.18 (Valuation rings are integrally closed proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A valuation ring V is integrally closed in its fraction field.

Solution. If x is integral over V but $x \notin V$, then $x^{-1} \in V$ and is a nonunit. Divide a monic equation for x by the highest power of x to express 1 as an element of the maximal ideal generated by powers of x^{-1} , contradiction. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 21.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Valuation rings impose a total divisibility order on a field. They are integrally closed local domains and provide a powerful way to test integrality and dominate local rings. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

21.13 Exercises with complete solutions

Exercise 21.20. State and apply the central principle behind Valuations. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A valuation assigns ordered-group values compatible with products and satisfying a triangle inequality for sums. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 21.21. State and apply the central principle behind Valuation rings. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A subring V of a field K is a valuation ring when for every nonzero $x \in K$, either x or x^{-1} lies in V . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 21.22. State and apply the central principle behind Locality. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The nonunits of a valuation ring form its unique maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 21.23. State and apply the central principle behind Divisibility order. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Principal ideals are totally ordered by inclusion in a valuation ring. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 21.24. State and apply the central principle behind Discrete valuation rings. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A DVR corresponds to a discrete rank-one valuation and has a principal maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 21.25. State and apply the central principle behind Integrally closed property. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Every valuation ring is integrally closed in its fraction field. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 21.26. State and apply the central principle behind Dominating local rings. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Valuation rings can contain a local domain while having maximal ideal contracting to the local maximal ideal. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 21.27. State and apply the central principle behind Prime ideals. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Prime ideals of a valuation ring are linearly ordered. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Part VI

Homological Algebra

Chapter 22

Chain Complexes

Chain complexes organize sequences of modules with consecutive compositions zero. Homology measures the failure of exactness and is the central invariant of homological algebra.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

22.1 Chain complexes

A chain complex has modules C_n and differentials $d_n : C_n \rightarrow C_{n-1}$ with $d_{n-1}d_n = 0$.

22.2 Cycles and boundaries

Cycles are kernels of differentials; boundaries are images of the next differentials.

22.3 Homology

$$H_n(C) = \ker d_n / \operatorname{im} d_{n+1}.$$

22.4 Exact complexes

A complex is exact exactly when all homology modules vanish.

22.5 Chain maps

A chain map commutes with differentials and therefore sends cycles to cycles and boundaries to boundaries.

22.6 Induced maps

Every chain map induces maps on homology.

22.7 Short exact sequences of complexes

Degree-wise short exact sequences produce long exact sequences in homology.

22.8 Homotopies

Chain-homotopic maps induce the same map on homology.

22.9 Examples

Two-term complexes, resolutions, Koszul-style complexes, and singular-like algebraic complexes illustrate the formalism.

22.10 Core structural results

Theorem 22.1 (Chain maps induce homology maps). *A chain map $f : C \rightarrow D$ induces $H_n(f) : H_n(C) \rightarrow H_n(D)$.*

Proof. Commutation with differentials sends cycles to cycles. It also sends boundaries to boundaries because $f_n d_{n+1} = d_{n+1} f_{n+1}$, so the map descends to the quotient. $\square$

Theorem 22.2 (Exactness equals zero homology). *A chain complex is exact at every degree iff $H_n(C) = 0$ for all n .*

Proof. By definition, zero homology at degree n means the cycle module equals the boundary module, which is exactly exactness at that term. $\square$

Theorem 22.3 (Chain homotopic maps agree on homology). *If $f - g = dh + hd$, then f and g induce the same homology maps.*

Proof. For a cycle z , $(f - g)(z) = dh(z)$ because $d(z) = 0$, so the difference of images is a boundary. $\square$

22.11 Worked examples

Example 22.4 (Chain complexes diagnostic). Give a rigorous worked analysis of Chain complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A chain complex has modules C_n and differentials $d_n : C_n \rightarrow C_{n-1}$ with $d_{n-1}d_n = 0$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 22.5 (Cycles and boundaries diagnostic). Give a rigorous worked analysis of Cycles and boundaries. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Cycles are kernels of differentials; boundaries are images of the next differentials. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 22.6 (Homology diagnostic). Give a rigorous worked analysis of Homology. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $H_n(C) = \ker d_n / \operatorname{im} d_{n+1}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 22.7 (Exact complexes diagnostic). Give a rigorous worked analysis of Exact complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A complex is exact exactly when all homology modules vanish. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

22.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 22.8 (Chain complexes diagnostic). Give a rigorous worked analysis of Chain complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A chain complex has modules C_n and differentials $d_n : C_n \rightarrow C_{n-1}$ with $d_{n-1}d_n = 0$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 22.9 (Cycles and boundaries diagnostic). Give a rigorous worked analysis of Cycles and boundaries. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Cycles are kernels of differentials; boundaries are images of the next differentials. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 22.10 (Homology diagnostic). Give a rigorous worked analysis of Homology. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $H_n(C) = \ker d_n / \operatorname{im} d_{n+1}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 22.11 (Exact complexes diagnostic). Give a rigorous worked analysis of Exact complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A complex is exact exactly when all homology modules vanish. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 22.12 (Chain maps diagnostic). Give a rigorous worked analysis of Chain maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A chain map commutes with differentials and therefore sends cycles to cycles and boundaries to boundaries. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 22.13 (Induced maps diagnostic). Give a rigorous worked analysis of Induced maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Every chain map induces maps on homology. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 22.14 (Short exact sequences of complexes diagnostic). Give a rigorous worked analysis of Short exact sequences of complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Degreewise short exact sequences produce long exact sequences in homology. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 22.15 (Homotopies diagnostic). Give a rigorous worked analysis of Homotopies. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Chain-homotopic maps induce the same map on homology. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 22.16 (Chain maps induce homology maps proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A chain map $f : C \rightarrow D$ induces $H_n(f) : H_n(C) \rightarrow H_n(D)$.

Solution. Commutation with differentials sends cycles to cycles. It also sends boundaries to boundaries because $f_n d_{n+1} = d_{n+1} f_{n+1}$, so the map descends to the quotient. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 22.17 (Exactness equals zero homology proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A chain complex is exact at every degree iff $H_n(C) = 0$ for all n .

Solution. By definition, zero homology at degree n means the cycle module equals the boundary module, which is exactly exactness at that term. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 22.18 (Chain homotopic maps agree on homology proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $f - g = dh + hd$, then f and g induce the same homology maps.

Solution. For a cycle z , $(f - g)(z) = dh(z)$ because $d(z) = 0$, so the difference of images is a boundary. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 22.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Chain complexes organize sequences of modules with consecutive compositions zero. Homology measures the failure of exactness and is the central invariant of homological algebra. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

22.13 Exercises with complete solutions

Exercise 22.20. State and apply the central principle behind Chain complexes. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A chain complex has modules C_n and differentials $d_n : C_n \rightarrow C_{n-1}$ with $d_{n-1}d_n = 0$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 22.21. State and apply the central principle behind Cycles and boundaries. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Cycles are kernels of differentials; boundaries are images of the next differentials. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 22.22. State and apply the central principle behind Homology. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $H_n(C) = \ker d_n / \operatorname{im} d_{n+1}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 22.23. State and apply the central principle behind Exact complexes. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A complex is exact exactly when all homology modules vanish. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 22.24. State and apply the central principle behind Chain maps. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A chain map commutes with differentials and therefore sends cycles to cycles and boundaries to boundaries. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 22.25. State and apply the central principle behind Induced maps. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Every chain map induces maps on homology. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 22.26. State and apply the central principle behind Short exact sequences of complexes. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Degreewise short exact sequences produce long exact sequences in homology. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 22.27. State and apply the central principle behind Homotopies. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Chain-homotopic maps induce the same map on homology. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 23

Free Resolutions

Free resolutions replace a module by an exact complex of free modules. They allow non-free modules to be studied using explicit matrices and are the computational input for Tor and Ext.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

23.1 Augmented complexes

A resolution ends in a surjection $F_0 \rightarrow M$ and is exact before and at M .

23.2 Free resolutions

All modules F_i in a free resolution are free.

23.3 Construction by kernels

Choose free generators for M , then free generators for the kernel, and repeat.

23.4 Projective resolutions

Projective resolutions generalize free resolutions and are often sufficient for derived functors.

23.5 Comparison maps

A module homomorphism lifts to a chain map between projective resolutions.

23.6 Homotopy uniqueness

Two such lifts are chain homotopic.

23.7 Length

Finite projective dimension means a resolution terminates after finitely many nonzero projectives.

23.8 Tensoring resolutions

Tensoring a free resolution computes Tor homology.

23.9 Applying Hom

Applying Hom to a projective resolution computes Ext cohomology.

23.10 Examples

Cyclic quotients and regular sequences yield explicit short or structured resolutions.

23.11 Core structural results

Theorem 23.1 (Existence of free resolutions). *Every module admits a free resolution.*

Proof. Choose a free module surjecting onto the module. Take its kernel, choose a free module surjecting onto that kernel, and iterate indefinitely. $\square$

Theorem 23.2 (Comparison theorem). *A homomorphism $M \rightarrow N$ lifts to a chain map from any projective resolution of M to any resolution of N ending in a surjection.*

Proof. Lift degree zero using projectivity. Inductively, the image of the previous differential lands in the next kernel by the chain condition, and projectivity lifts through the corresponding surjection. $\square$

Theorem 23.3 (Homotopy uniqueness of comparison maps). *Two chain maps between projective resolutions inducing the same module map are chain homotopic.*

Proof. Construct the homotopy inductively. At each degree, the difference of the two maps has image in a kernel already identified with the image of the next differential, and projectivity provides a lift. $\square$

23.12 Worked examples

Example 23.4 (Augmented complexes diagnostic). Give a rigorous worked analysis of Augmented complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A resolution ends in a surjection $F_0 \rightarrow M$ and is exact before and at M . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 23.5 (Free resolutions diagnostic). Give a rigorous worked analysis of Free resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. All modules F_i in a free resolution are free. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 23.6 (Construction by kernels diagnostic). Give a rigorous worked analysis of Construction by kernels. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Choose free generators for M , then free generators for the kernel, and repeat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 23.7 (Projective resolutions diagnostic). Give a rigorous worked analysis of Projective resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Projective resolutions generalize free resolutions and are often sufficient for derived functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

23.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 23.8 (Augmented complexes diagnostic). Give a rigorous worked analysis of Augmented complexes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A resolution ends in a surjection $F_0 \rightarrow M$ and is exact before and at M . The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 23.9 (Free resolutions diagnostic). Give a rigorous worked analysis of Free resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. All modules F_i in a free resolution are free. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 23.10 (Construction by kernels diagnostic). Give a rigorous worked analysis of Construction by kernels. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Choose free generators for M , then free generators for the kernel, and repeat. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 23.11 (Projective resolutions diagnostic). Give a rigorous worked analysis of Projective resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Projective resolutions generalize free resolutions and are often sufficient for derived functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 23.12 (Comparison maps diagnostic). Give a rigorous worked analysis of Comparison maps. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module homomorphism lifts to a chain map between projective resolutions. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 23.13 (Homotopy uniqueness diagnostic). Give a rigorous worked analysis of Homotopy uniqueness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Two such lifts are chain homotopic. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 23.14 (Length diagnostic). Give a rigorous worked analysis of Length. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Finite projective dimension means a resolution terminates after finitely many nonzero projectives. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 23.15 (Tensoring resolutions diagnostic). Give a rigorous worked analysis of Tensoring resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Tensoring a free resolution computes Tor homology. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 23.16 (Existence of free resolutions proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Every module admits a free resolution.

Solution. Choose a free module surjecting onto the module. Take its kernel, choose a free module surjecting onto that kernel, and iterate indefinitely. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 23.17 (Comparison theorem proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A homomorphism $M \rightarrow N$ lifts to a chain map from any projective resolution of M to any resolution of N ending in a surjection.

Solution. Lift degree zero using projectivity. Inductively, the image of the previous differential lands in the next kernel by the chain condition, and projectivity lifts through the corresponding surjection. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 23.18 (Homotopy uniqueness of comparison maps proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Two chain maps between projective resolutions inducing the same module map are chain homotopic.

Solution. Construct the homotopy inductively. At each degree, the difference of the two maps has image in a kernel already identified with the image of the next differential, and projectivity provides a lift. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 23.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Free resolutions replace a module by an exact complex of free modules. They allow non-free modules to be studied using explicit matrices and are the computational input for Tor and Ext. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

23.14 Exercises with complete solutions

Exercise 23.20. State and apply the central principle behind Augmented complexes. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A resolution ends in a surjection $F_0 \rightarrow M$ and is exact before and at M . The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 23.21. State and apply the central principle behind Free resolutions. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. All modules F_i in a free resolution are free. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 23.22. State and apply the central principle behind Construction by kernels. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Choose free generators for M , then free generators for the kernel, and repeat. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 23.23. State and apply the central principle behind Projective resolutions. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Projective resolutions generalize free resolutions and are often sufficient for derived functors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 23.24. State and apply the central principle behind Comparison maps. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A module homomorphism lifts to a chain map between projective resolutions. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 23.25. State and apply the central principle behind Homotopy uniqueness. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Two such lifts are chain homotopic. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 23.26. State and apply the central principle behind Length. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Finite projective dimension means a resolution terminates after finitely many nonzero projectives. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 23.27. State and apply the central principle behind Tensoring resolutions. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Tensoring a free resolution computes Tor homology. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 24

Syzygies

Syzygies are relations among generators, then relations among relations, and so on. Iterating syzygies turns presentations into resolutions and exposes hidden structure in modules.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

24.1 First syzygy

Given a surjection $F_0 \rightarrow M$, the first syzygy module is its kernel.

24.2 Higher syzygies

The n th syzygy is the kernel at stage $n - 1$ of a chosen free or projective resolution.

24.3 Dependence on presentation

Syzygies depend on choices, but stable projective information is controlled up to adding free summands.

24.4 Presentations

A finite presentation gives a finitely generated first syzygy.

24.5 Noetherian setting

Over a Noetherian ring, syzygies of finite modules in finite free presentations remain finitely generated.

24.6 Dimension shifting

Higher Tor and Ext groups can be expressed through lower groups of syzygies.

24.7 Regular sequences

Koszul-type relations provide canonical syzygies for complete-intersection situations.

24.8 Hilbert syzygy preview

Over polynomial rings over fields, finite modules have bounded projective resolution length.

24.9 Computational matrices

Syzygies are kernels of matrices and can be studied algorithmically.

24.10 Core structural results

Theorem 24.1 (Syzygies over Noetherian rings are finite). *If A is Noetherian and M is finitely generated, then the kernel of a map $A^n \rightarrow M$ is finitely generated.*

Proof. The finite free module A^n is Noetherian, so every submodule, including the kernel, is finitely generated. $\square$

Theorem 24.2 (Schanuel lemma). *If $0 \rightarrow K \rightarrow P \rightarrow M \rightarrow 0$ and $0 \rightarrow K' \rightarrow P' \rightarrow M \rightarrow 0$ have projective P, P' , then $K \oplus P' \cong K' \oplus P$.*

Proof. Form the pullback of $P \rightarrow M$ and $P' \rightarrow M$. Its two projections fit into short exact sequences that split because P and P' are projective, yielding the two decompositions. $\square$

Theorem 24.3 (Dimension shifting for Tor). *If $0 \rightarrow \Omega M \rightarrow P \rightarrow M \rightarrow 0$ with P projective, then $\operatorname{Tor}_n^A(M, N) \cong \operatorname{Tor}_{n-1}^A(\Omega M, N)$ for $n \geq 2$.*

Proof. Apply the long exact Tor sequence. Higher Tor of the projective module P vanishes, so adjacent terms give the stated isomorphism. $\square$

24.11 Worked examples

Example 24.4 (First syzygy diagnostic). Give a rigorous worked analysis of First syzygy. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Given a surjection $F_0 \rightarrow M$, the first syzygy module is its kernel. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 24.5 (Higher syzygies diagnostic). Give a rigorous worked analysis of Higher syzygies. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. The n th syzygy is the kernel at stage $n - 1$ of a chosen free or projective resolution. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 24.6 (Dependence on presentation diagnostic). Give a rigorous worked analysis of Dependence on presentation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Syzygies depend on choices, but stable projective information is controlled up to adding free summands. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 24.7 (Presentations diagnostic). Give a rigorous worked analysis of Presentations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A finite presentation gives a finitely generated first syzygy. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

24.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 24.8 (First syzygy diagnostic). Give a rigorous worked analysis of First syzygy. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Given a surjection $F_0 \rightarrow M$, the first syzygy module is its kernel. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 24.9 (Higher syzygies diagnostic). Give a rigorous worked analysis of Higher syzygies. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The n th syzygy is the kernel at stage $n - 1$ of a chosen free or projective resolution. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 24.10 (Dependence on presentation diagnostic). Give a rigorous worked analysis of Dependence on presentation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Syzygies depend on choices, but stable projective information is controlled up to adding free summands. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 24.11 (Presentations diagnostic). Give a rigorous worked analysis of Presentations. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A finite presentation gives a finitely generated first syzygy. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 24.12 (Noetherian setting diagnostic). Give a rigorous worked analysis of Noetherian setting. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over a Noetherian ring, syzygies of finite modules in finite free presentations remain finitely generated. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 24.13 (Dimension shifting diagnostic). Give a rigorous worked analysis of Dimension shifting. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Higher Tor and Ext groups can be expressed through lower groups of syzygies. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 24.14 (Regular sequences diagnostic). Give a rigorous worked analysis of Regular sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Koszul-type relations provide canonical syzygies for complete-intersection situations. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 24.15 (Hilbert syzygy preview diagnostic). Give a rigorous worked analysis of Hilbert syzygy preview. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over polynomial rings over fields, finite modules have bounded projective resolution length. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 24.16 (Syzygies over Noetherian rings are finite proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If A is Noetherian and M is finitely generated, then the kernel of a map $A^n \rightarrow M$ is finitely generated.

Solution. The finite free module A^n is Noetherian, so every submodule, including the kernel, is finitely generated. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 24.17 (Schanuel lemma proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $0 \rightarrow K \rightarrow P \rightarrow M \rightarrow 0$ and $0 \rightarrow K' \rightarrow P' \rightarrow M \rightarrow 0$ have projective P, P' , then $K \oplus P' \cong K' \oplus P$.

Solution. Form the pullback of $P \rightarrow M$ and $P' \rightarrow M$. Its two projections fit into short exact sequences that split because P and P' are projective, yielding the two decompositions. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 24.18 (Dimension shifting for Tor proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If $0 \rightarrow \Omega M \rightarrow P \rightarrow M \rightarrow 0$ with P projective, then $\operatorname{Tor}_n^A(M, N) \cong \operatorname{Tor}_{n-1}^A(\Omega M, N)$ for $n \geq 2$.

Solution. Apply the long exact Tor sequence. Higher Tor of the projective module P vanishes, so adjacent terms give the stated isomorphism. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 24.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Syzygies are relations among generators, then relations among relations, and so on. Iterating syzygies turns presentations into resolutions and exposes hidden structure in modules. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

24.13 Exercises with complete solutions

Exercise 24.20. State and apply the central principle behind First syzygy. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Given a surjection $F_0 \rightarrow M$, the first syzygy module is its kernel. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 24.21. State and apply the central principle behind Higher syzygies. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The n th syzygy is the kernel at stage $n - 1$ of a chosen free or projective resolution. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 24.22. State and apply the central principle behind Dependence on presentation. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Syzygies depend on choices, but stable projective information is controlled up to adding free summands. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 24.23. State and apply the central principle behind Presentations. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A finite presentation gives a finitely generated first syzygy. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 24.24. State and apply the central principle behind Noetherian setting. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Over a Noetherian ring, syzygies of finite modules in finite free presentations remain finitely generated. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 24.25. State and apply the central principle behind Dimension shifting. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Higher Tor and Ext groups can be expressed through lower groups of syzygies. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 24.26. State and apply the central principle behind Regular sequences. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Koszul-type relations provide canonical syzygies for complete-intersection situations. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 24.27. State and apply the central principle behind Hilbert syzygy preview. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Over polynomial rings over fields, finite modules have bounded projective resolution length. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 25

Minimal Resolutions

Minimal resolutions remove redundant free summands. Over local rings, minimality is detected by differentials whose matrix entries lie in the maximal ideal, making Betti numbers intrinsic.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

25.1 Minimal free resolutions

A free resolution is minimal when no contractible free summand can be removed.

25.2 Local criterion

Over a local ring $(A, \mathfrak{m})$, a finite free resolution is minimal iff each differential has image contained in $\mathfrak{m}$ times the previous free module.

25.3 Reduction modulo the maximal ideal

Minimal differentials become zero after tensoring with the residue field.

25.4 Betti numbers

Ranks of the free modules in a minimal resolution are invariants of the module.

25.5 Nakayama mechanism

Minimal generating sets are detected after reduction modulo $\mathfrak{m}$.

25.6 Uniqueness

Minimal free resolutions over local rings are unique up to chain isomorphism.

25.7 Projective dimension

The last nonzero module in a minimal finite resolution detects projective dimension.

25.8 Graded analogue

Over standard graded rings, minimality is similarly controlled by positive-degree entries.

25.9 Computational significance

Minimal resolutions expose generators, first relations, and higher relations without algebraic cancellation.

25.10 Core structural results

Theorem 25.1 (Local minimality criterion). *A free resolution $F_\bullet$ over a local ring is minimal iff $d(F_n) \subseteq \mathfrak{m}F_{n-1}$ for all n .*

Proof. If a matrix entry is a unit, elementary basis changes split off an identity map between rank-one free summands, producing a contractible piece. Conversely, if all entries lie in the maximal ideal, no such unit cancellation is possible. $\square$

Theorem 25.2 (Betti numbers from Tor). *For a minimal free resolution of a finite module over a local ring with residue field k , $\mathrm{Tor}_n^A(M, k) \cong F_n \otimes_A k$.*

Proof. Tensor the minimal resolution with k . Every differential becomes zero because its entries lie in $\mathfrak{m}$, so homology in degree n is the entire vector space $F_n \otimes k$. $\square$

Theorem 25.3 (Uniqueness of minimal resolutions). *Minimal free resolutions of a finite module over a local ring are isomorphic as complexes.*

Proof. Comparison maps in both directions lift the identity of the module. Modulo the maximal ideal they are inverse isomorphisms on Tor, hence degreewise isomorphisms by Nakayama's lemma. $\square$

25.11 Worked examples

Example 25.4 (Minimal free resolutions diagnostic). Give a rigorous worked analysis of Minimal free resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. A free resolution is minimal when no contractible free summand can be removed. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 25.5 (Local criterion diagnostic). Give a rigorous worked analysis of Local criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Over a local ring $(A, \mathfrak{m})$, a finite free resolution is minimal iff each

differential has image contained in $\mathfrak{m}$ times the previous free module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 25.6 (Reduction modulo the maximal ideal diagnostic). Give a rigorous worked analysis of Reduction modulo the maximal ideal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Minimal differentials become zero after tensoring with the residue field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 25.7 (Betti numbers diagnostic). Give a rigorous worked analysis of Betti numbers. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Ranks of the free modules in a minimal resolution are invariants of the module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

25.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 25.8 (Minimal free resolutions diagnostic). Give a rigorous worked analysis of Minimal free resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A free resolution is minimal when no contractible free summand can be removed. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 25.9 (Local criterion diagnostic). Give a rigorous worked analysis of Local criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over a local ring $(A, \mathfrak{m})$, a finite free resolution is minimal iff each differential has image contained in $\mathfrak{m}$ times the previous free module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 25.10 (Reduction modulo the maximal ideal diagnostic). Give a rigorous worked analysis of Reduction modulo the maximal ideal. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Minimal differentials become zero after tensoring with the residue field. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 25.11 (Betti numbers diagnostic). Give a rigorous worked analysis of Betti numbers. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ranks of the free modules in a minimal resolution are invariants of the module. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 25.12 (Nakayama mechanism diagnostic). Give a rigorous worked analysis of Nakayama mechanism. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Minimal generating sets are detected after reduction modulo $\mathfrak{m}$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 25.13 (Uniqueness diagnostic). Give a rigorous worked analysis of Uniqueness. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Minimal free resolutions over local rings are unique up to chain isomorphism. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 25.14 (Projective dimension diagnostic). Give a rigorous worked analysis of Projective dimension. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. The last nonzero module in a minimal finite resolution detects projective dimension. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 25.15 (Graded analogue diagnostic). Give a rigorous worked analysis of Graded analogue. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over standard graded rings, minimality is similarly controlled by positive-degree entries. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 25.16 (Local minimality criterion proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A free resolution $F_\bullet$ over a local ring is minimal iff $d(F_n) \subseteq \mathfrak{m}F_{n-1}$ for all n .

Solution. If a matrix entry is a unit, elementary basis changes split off an identity map between rank-one free summands, producing a contractible piece. Conversely, if all entries lie in the maximal ideal, no such unit cancellation is possible. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 25.17 (Betti numbers from Tor proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For a minimal free resolution of a finite module over a local ring with residue field k , $\mathrm{Tor}_n^A(M, k) \cong F_n \otimes_A k$.

Solution. Tensor the minimal resolution with k . Every differential becomes zero because its entries lie in $\mathfrak{m}$, so homology in degree n is the entire vector space $F_n \otimes k$. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 25.18 (Uniqueness of minimal resolutions proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Minimal free resolutions of a finite module over a local ring are isomorphic as complexes.

Solution. Comparison maps in both directions lift the identity of the module. Modulo the maximal ideal they are inverse isomorphisms on Tor , hence degreewise isomorphisms by Nakayama's lemma. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 25.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Minimal resolutions remove redundant free summands. Over local rings, minimality is detected by differentials whose matrix entries lie in the maximal ideal, making Betti numbers intrinsic. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

25.13 Exercises with complete solutions

Exercise 25.20. State and apply the central principle behind Minimal free resolutions. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A free resolution is minimal when no contractible free summand can be removed. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 25.21. State and apply the central principle behind Local criterion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Over a local ring $(A, \mathfrak{m})$, a finite free resolution is minimal iff each differential has image contained in $\mathfrak{m}$ times the previous free module. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 25.22. State and apply the central principle behind Reduction modulo the maximal ideal. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Minimal differentials become zero after tensoring with the residue field. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 25.23. State and apply the central principle behind Betti numbers. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Ranks of the free modules in a minimal resolution are invariants of the module. The same principle can then be reused in later localization, support, base-change, resolution, Tor , or Ext arguments. $\square$

Exercise 25.24. State and apply the central principle behind Nakayama mechanism. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Minimal generating sets are detected after reduction modulo $\mathfrak{m}$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 25.25. State and apply the central principle behind Uniqueness. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Minimal free resolutions over local rings are unique up to chain isomorphism. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 25.26. State and apply the central principle behind Projective dimension. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. The last nonzero module in a minimal finite resolution detects projective dimension. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 25.27. State and apply the central principle behind Graded analogue. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Over standard graded rings, minimality is similarly controlled by positive-degree entries. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 26

The Tor Functor

The Tor functor is the homology left behind when tensor product fails to be exact. It is computed from projective resolutions and detects torsion, non-flatness, and derived intersections.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

26.1 Definition

Choose a projective resolution $P_\bullet \rightarrow M$ and set $\mathrm{Tor}_n^A(M, N) = H_n(P_\bullet \otimes_A N)$.

26.2 Independence of resolution

Comparison and homotopy theorems make Tor well-defined up to canonical isomorphism.

26.3 Degree zero

$\mathrm{Tor}_0^A(M, N) \cong M \otimes_A N$.

26.4 Long exact sequence

Short exact sequences in either variable yield long exact Tor sequences.

26.5 Projective vanishing

Higher Tor with a projective module vanishes.

26.6 Flatness criterion

A module is flat iff $\mathrm{Tor}_1^A(-, F) = 0$ for all modules.

26.7 Symmetry

Over a commutative ring, Tor is symmetric in its two module arguments.

26.8 Cyclic calculation

For integers or quotient modules, short resolutions compute Tor explicitly.

26.9 Dimension shifting

Syzygies reduce higher Tor computations to lower degrees.

26.10 Core structural results

Theorem 26.1 (Well-definedness of Tor). *The modules $H_n(P_\bullet \otimes N)$ are independent of the chosen projective resolution of M up to canonical isomorphism.*

Proof. Comparison maps between projective resolutions are unique up to chain homotopy. Tensor preserves chain homotopies, and chain-homotopic maps induce the same homology maps, yielding canonical isomorphisms. $\square$

Theorem 26.2 (Projective vanishing). *If P is projective, then $\mathrm{Tor}_n^A(P, N) = 0$ for $n > 0$.*

Proof. Use the resolution concentrated in degree zero at P . The tensor complex has no positive-degree terms, so its positive homology vanishes. $\square$

Theorem 26.3 (Flatness via Tor). *An A -module F is flat iff $\mathrm{Tor}_1^A(A/I, F) = 0$ for every finitely generated ideal I under the standard finite-ideal criterion, and in particular iff $\mathrm{Tor}_1^A(M, F) = 0$ for all M .*

Proof. Apply Tor to $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$. Vanishing of the connecting kernel is exactly injectivity of $I \otimes F \rightarrow F$; the general criterion follows from the flatness tests and long exact sequences. $\square$

26.11 Worked examples

Example 26.4 (Definition diagnostic). Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Choose a projective resolution $P_\bullet \rightarrow M$ and set $\mathrm{Tor}_n^A(M, N) = H_n(P_\bullet \otimes_A N)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 26.5 (Independence of resolution diagnostic). Give a rigorous worked analysis of Independence of resolution. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Comparison and homotopy theorems make Tor well-defined up to canonical isomorphism. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 26.6 (Degree zero diagnostic). Give a rigorous worked analysis of Degree zero. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\mathrm{Tor}_0^A(M, N) \cong M \otimes_A N$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 26.7 (Long exact sequence diagnostic). Give a rigorous worked analysis of Long exact sequence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Short exact sequences in either variable yield long exact Tor sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

26.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 26.8 (Definition diagnostic). Give a rigorous worked analysis of Definition. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Choose a projective resolution $P_\bullet \rightarrow M$ and set $\mathrm{Tor}_n^A(M, N) = H_n(P_\bullet \otimes_A N)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 26.9 (Independence of resolution diagnostic). Give a rigorous worked analysis of Independence of resolution. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Comparison and homotopy theorems make Tor well-defined up to canonical isomorphism. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 26.10 (Degree zero diagnostic). Give a rigorous worked analysis of Degree zero. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\mathrm{Tor}_0^A(M, N) \cong M \otimes_A N$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 26.11 (Long exact sequence diagnostic). Give a rigorous worked analysis of Long exact sequence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Short exact sequences in either variable yield long exact Tor sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 26.12 (Projective vanishing diagnostic). Give a rigorous worked analysis of Projective vanishing. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Higher Tor with a projective module vanishes. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 26.13 (Flatness criterion diagnostic). Give a rigorous worked analysis of Flatness criterion. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. A module is flat iff $\mathrm{Tor}_1^A(-, F) = 0$ for all modules. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 26.14 (Symmetry diagnostic). Give a rigorous worked analysis of Symmetry. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Over a commutative ring, Tor is symmetric in its two module arguments. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 26.15 (Cyclic calculation diagnostic). Give a rigorous worked analysis of Cyclic calculation. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. For integers or quotient modules, short resolutions compute Tor explicitly. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 26.16 (Well-definedness of Tor proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: The modules $H_n(P_\bullet \otimes N)$ are independent of the chosen projective resolution of M up to canonical isomorphism.

Solution. Comparison maps between projective resolutions are unique up to chain homotopy. Tensor preserves chain homotopies, and chain-homotopic maps induce the same homology maps, yielding canonical isomorphisms. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 26.17 (Projective vanishing proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If P is projective, then $\mathrm{Tor}_n^A(P, N) = 0$ for $n > 0$.

Solution. Use the resolution concentrated in degree zero at P . The tensor complex has no positive-degree terms, so its positive homology vanishes. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 26.18 (Flatness via Tor proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: An A -module F is flat iff $\mathrm{Tor}_1^A(A/I, F) = 0$ for every finitely generated ideal I under the standard finite-ideal criterion, and in particular iff $\mathrm{Tor}_1^A(M, F) = 0$ for all M .

Solution. Apply Tor to $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$. Vanishing of the connecting kernel is exactly injectivity of $I \otimes F \rightarrow F$; the general criterion follows from the flatness tests and long exact sequences. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 26.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. The Tor functor is the homology left behind when tensor product fails to be exact. It is computed from projective resolutions and detects torsion, non-flatness, and derived intersections. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

26.13 Exercises with complete solutions

Exercise 26.20. State and apply the central principle behind Definition. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Choose a projective resolution $P_\bullet \rightarrow M$ and set $\mathrm{Tor}_n^A(M, N) = H_n(P_\bullet \otimes_A N)$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 26.21. State and apply the central principle behind Independence of resolution. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Comparison and homotopy theorems make Tor well-defined up to canonical isomorphism. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 26.22. State and apply the central principle behind Degree zero. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\mathrm{Tor}_0^A(M, N) \cong M \otimes_A N$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 26.23. State and apply the central principle behind Long exact sequence. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Short exact sequences in either variable yield long exact Tor sequences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 26.24. State and apply the central principle behind Projective vanishing. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Higher Tor with a projective module vanishes. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 26.25. State and apply the central principle behind Flatness criterion. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. A module is flat iff $\mathrm{Tor}_1^A(-, F) = 0$ for all modules. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 26.26. State and apply the central principle behind Symmetry. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Over a commutative ring, Tor is symmetric in its two module arguments. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 26.27. State and apply the central principle behind Cyclic calculation. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. For integers or quotient modules, short resolutions compute Tor explicitly. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 27

The Ext Functor

The Ext functor is the derived correction to the left exact Hom functor. It classifies extensions in degree one and measures higher obstructions to lifting maps.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

27.1 Definition from projectives

Choose a projective resolution $P_\bullet \rightarrow M$ and set $\text{Ext}_A^n(M, N) = H^n(\text{Hom}_A(P_\bullet, N))$.

27.2 Degree zero

$\text{Ext}_A^0(M, N) \cong \text{Hom}_A(M, N)$.

27.3 Long exact sequence

Short exact sequences yield long exact Ext sequences.

27.4 Projective vanishing

Higher Ext out of a projective module is zero.

27.5 Injective viewpoint

Ext can also be computed using injective resolutions in the second variable.

27.6 Extension classes

$\text{Ext}^1(M, N)$ classifies short exact sequences $0 \rightarrow N \rightarrow E \rightarrow M \rightarrow 0$ up to the appropriate equivalence.

27.7 Dimension shifting

Syzygies reduce higher Ext to lower Ext.

27.8 Local and graded variants

Localization and graded structures refine Ext in common Noetherian settings.

27.9 Duality preview

Ext groups appear in duality, canonical modules, and deformation theory.

27.10 Core structural results

Theorem 27.1 (Well-definedness of Ext). *$H^n(\text{Hom}(P_\bullet, N))$ is independent of the chosen projective resolution of M .*

Proof. Comparison maps between resolutions are unique up to chain homotopy. Applying Hom turns chain homotopies into cochain homotopies, which induce identical cohomology maps. $\square$

Theorem 27.2 (Projective vanishing). *If P is projective, then $\text{Ext}_A^n(P, N) = 0$ for $n > 0$.*

Proof. Use the projective resolution concentrated in degree zero. The resulting Hom cochain complex has no positive cohomology. $\square$

Theorem 27.3 (Ext one classifies extensions). *Equivalence classes of short exact sequences $0 \rightarrow N \rightarrow E \rightarrow M \rightarrow 0$ form a group naturally isomorphic to $\text{Ext}_A^1(M, N)$.*

Proof. Pull an extension back along a projective presentation of M and choose a splitting over the projective term. The resulting map from the first syzygy to N defines a degree-one cohomology class; changing the splitting changes it by a coboundary, and the construction is reversible via pushout. $\square$

27.11 Worked examples

Example 27.4 (Definition from projectives diagnostic). Give a rigorous worked analysis of Definition from projectives. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Choose a projective resolution $P_\bullet \rightarrow M$ and set $\text{Ext}_A^n(M, N) = H^n(\text{Hom}_A(P_\bullet, N))$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 27.5 (Degree zero diagnostic). Give a rigorous worked analysis of Degree zero. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. $\text{Ext}_A^0(M, N) \cong \text{Hom}_A(M, N)$. The decisive step is to use the universal

property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 27.6 (Long exact sequence diagnostic). Give a rigorous worked analysis of Long exact sequence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Short exact sequences yield long exact Ext sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 27.7 (Projective vanishing diagnostic). Give a rigorous worked analysis of Projective vanishing. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Higher Ext out of a projective module is zero. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

27.12 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 27.8 (Definition from projectives diagnostic). Give a rigorous worked analysis of Definition from projectives. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Choose a projective resolution $P_\bullet \rightarrow M$ and set $\text{Ext}_A^n(M, N) = H^n(\text{Hom}_A(P_\bullet, N))$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 27.9 (Degree zero diagnostic). Give a rigorous worked analysis of Degree zero. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\text{Ext}_A^0(M, N) \cong \text{Hom}_A(M, N)$. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 27.10 (Long exact sequence diagnostic). Give a rigorous worked analysis of Long exact sequence. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Short exact sequences yield long exact Ext sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 27.11 (Projective vanishing diagnostic). Give a rigorous worked analysis of Projective vanishing. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Higher Ext out of a projective module is zero. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 27.12 (Injective viewpoint diagnostic). Give a rigorous worked analysis of Injective viewpoint. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ext can also be computed using injective resolutions in the second variable. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 27.13 (Extension classes diagnostic). Give a rigorous worked analysis of Extension classes. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. $\text{Ext}^1(M, N)$ classifies short exact sequences $0 \rightarrow N \rightarrow E \rightarrow M \rightarrow 0$ up to the appropriate equivalence. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 27.14 (Dimension shifting diagnostic). Give a rigorous worked analysis of Dimension shifting. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Syzygies reduce higher Ext to lower Ext. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 27.15 (Local and graded variants diagnostic). Give a rigorous worked analysis of Local and graded variants. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Localization and graded structures refine Ext in common Noetherian settings. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 27.16 (Well-definedness of Ext proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: $H^n(\text{Hom}(P_\bullet, N))$ is independent of the chosen projective resolution of M .

Solution. Comparison maps between resolutions are unique up to chain homotopy. Applying Hom turns chain homotopies into cochain homotopies, which induce identical cohomology maps. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 27.17 (Projective vanishing proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: If P is projective, then $\text{Ext}_A^n(P, N) = 0$ for $n > 0$.

Solution. Use the projective resolution concentrated in degree zero. The resulting Hom cochain complex has no positive cohomology. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 27.18 (Ext one classifies extensions proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: Equivalence classes of short exact sequences $0 \rightarrow N \rightarrow E \rightarrow M \rightarrow 0$ form a group naturally isomorphic to $\text{Ext}_A^1(M, N)$.

Solution. Pull an extension back along a projective presentation of M and choose a splitting over the projective term. The resulting map from the first syzygy to N defines a degree-one cohomology class; changing the splitting changes it by a coboundary, and the construction is reversible via pushout. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 27.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. The Ext functor is the derived correction to the left exact Hom functor. It classifies extensions in degree one and measures higher obstructions to lifting maps. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

27.13 Exercises with complete solutions

Exercise 27.20. State and apply the central principle behind Definition from projectives. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Choose a projective resolution $P_\bullet \rightarrow M$ and set $\text{Ext}_A^n(M, N) = H^n(\text{Hom}_A(P_\bullet, N))$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 27.21. State and apply the central principle behind Degree zero. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\text{Ext}_A^0(M, N) \cong \text{Hom}_A(M, N)$. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 27.22. State and apply the central principle behind Long exact sequence. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Short exact sequences yield long exact Ext sequences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 27.23. State and apply the central principle behind Projective vanishing. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Higher Ext out of a projective module is zero. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 27.24. State and apply the central principle behind Injective viewpoint. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Ext can also be computed using injective resolutions in the second variable. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 27.25. State and apply the central principle behind Extension classes. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. $\text{Ext}^1(M, N)$ classifies short exact sequences $0 \rightarrow N \rightarrow E \rightarrow M \rightarrow 0$ up to the appropriate equivalence. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 27.26. State and apply the central principle behind Dimension shifting. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Syzygies reduce higher Ext to lower Ext. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 27.27. State and apply the central principle behind Local and graded variants. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Localization and graded structures refine Ext in common Noetherian settings. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.

Chapter 28

Derived-Functor Viewpoint

Derived functors organize Tor, Ext, and similar constructions as systematic corrections to nonexact functors. Resolutions make these corrections computable, while long exact sequences expose their functorial structure.

Learning goals

The reader should be able to state the definitions precisely, prove the core structural results, move between global and localized formulations, and audit finiteness and exactness hypotheses.

Conceptual roadmap

presentation or universal property $\longrightarrow$ exactness/localization $\longrightarrow$ structural consequence.

28.1 Exact and derived behavior

Left or right exact functors lose information at kernels or cokernels; derived functors measure that failure in higher degrees.

28.2 Left derived functors

Projective or flat resolutions produce left derived functors of right exact functors.

28.3 Right derived functors

Injective resolutions produce right derived functors of left exact functors.

28.4 Tor as derived tensor

Tor_n is the left derived functor of tensor product.

28.5 Ext as derived Hom

Ext^n is the right derived functor of Hom in the second variable or equivalently computed projectively in the first.

28.6 Universality

Derived functors are characterized by effaceability and universal delta-functor properties.

28.7 Long exact sequences

Connecting morphisms package the failure of exactness into functorial long exact sequences.

28.8 Acyclic resolutions

One may compute derived functors using suitable classes of acyclic objects, not only projectives or injectives.

28.9 Derived-category preview

Complexes and quasi-isomorphisms provide a setting in which derived functors become ordinary functors after localization.

28.10 Algebra-geometry bridge

Derived tensor and derived Hom underpin intersection theory, deformation theory, and sheaf cohomology.

28.11 Core structural results

Theorem 28.1 (Tor is the left derived tensor functor). *For fixed N , the homology of a projective resolution tensor N gives the left derived functors of $- \otimes_A N$.*

Proof. Projective resolutions are acyclic for tensor in positive resolution degrees and comparison maps make the construction functorial and independent of resolution; degree zero recovers ordinary tensor. $\square$

Theorem 28.2 (Ext is a derived Hom functor). *For fixed N , cohomology of $\text{Hom}(P_\bullet, N)$ gives derived functors extending $\text{Hom}(-, N)$ contravariantly in the first variable.*

Proof. Projective resolutions replace the first variable by Hom-acyclic objects. Comparison and homotopy yield functorial well-defined cohomology, and degree zero recovers Hom. $\square$

Theorem 28.3 (Derived functors produce long exact sequences). *A short exact sequence in the appropriate variable induces a natural long exact sequence of derived functors.*

Proof. Resolve compatibly or apply the functor to a short exact sequence of suitable resolutions. The resulting short exact sequence of complexes yields the standard long exact sequence in homology or cohomology via connecting morphisms. $\square$

28.12 Worked examples

Example 28.4 (Exact and derived behavior diagnostic). Give a rigorous worked analysis of Exact and derived behavior. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Left or right exact functors lose information at

kernels or cokernels; derived functors measure that failure in higher degrees. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 28.5 (Left derived functors diagnostic). Give a rigorous worked analysis of Left derived functors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Projective or flat resolutions produce left derived functors of right exact functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 28.6 (Right derived functors diagnostic). Give a rigorous worked analysis of Right derived functors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Injective resolutions produce right derived functors of left exact functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

Example 28.7 (Tor as derived tensor diagnostic). Give a rigorous worked analysis of Tor as derived tensor. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion. Tor_n is the left derived functor of tensor product. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure.

28.13 Solved dossiers

Each dossier is a canonical solved problem. The provenance ledger records which retained corpus rules guided its topic and which dossiers were added freshly to complete the chapter.

Problem 28.8 (Exact and derived behavior diagnostic). Give a rigorous worked analysis of Exact and derived behavior. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Left or right exact functors lose information at kernels or cokernels; derived functors measure that failure in higher degrees. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 28.9 (Left derived functors diagnostic). Give a rigorous worked analysis of Left derived functors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Projective or flat resolutions produce left derived functors of right exact functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 28.10 (Right derived functors diagnostic). Give a rigorous worked analysis of Right derived functors. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Injective resolutions produce right derived functors of left exact functors. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 28.11 (Tor as derived tensor diagnostic). Give a rigorous worked analysis of Tor as derived tensor. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Tor_n is the left derived functor of tensor product. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 28.12 (Ext as derived Hom diagnostic). Give a rigorous worked analysis of Ext as derived Hom. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Ext^n is the right derived functor of Hom in the second variable or equivalently computed projectively in the first. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 28.13 (Universality diagnostic). Give a rigorous worked analysis of Universality. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Derived functors are characterized by effaceability and universal delta-functor properties. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 28.14 (Long exact sequences diagnostic). Give a rigorous worked analysis of Long exact sequences. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. Connecting morphisms package the failure of exactness into functorial long exact sequences. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 28.15 (Acyclic resolutions diagnostic). Give a rigorous worked analysis of Acyclic resolutions. State the ring, module, finiteness, localization, or homological hypotheses and derive the central conclusion.

Solution. One may compute derived functors using suitable classes of acyclic objects, not only projectives or injectives. The decisive step is to use the universal property, exactness criterion, localization test, or finite-generation mechanism appropriate to the algebraic structure. $\square$

Problem 28.16 (Tor is the left derived tensor functor proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For fixed N , the homology of a projective resolution tensor N gives the left derived functors of $- \otimes_A N$.

Solution. Projective resolutions are acyclic for tensor in positive resolution degrees and comparison maps make the construction functorial and independent of resolution; degree zero recovers ordinary tensor. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 28.17 (Ext is a derived Hom functor proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: For fixed N , cohomology of $\text{Hom}(P_\bullet, N)$ gives derived functors extending $\text{Hom}(-, N)$ contravariantly in the first variable.

Solution. Projective resolutions replace the first variable by Hom-acyclic objects. Comparison and homotopy yield functorial well-defined cohomology, and degree zero recovers Hom. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 28.18 (Derived functors produce long exact sequences proof dossier). Prove the following structural statement and identify the hypothesis that makes the conclusion work: A short exact sequence in the appropriate variable induces a natural long exact sequence of derived functors.

Solution. Resolve compatibly or apply the functor to a short exact sequence of suitable resolutions. The resulting short exact sequence of complexes yields the standard long exact sequence in homology or cohomology via connecting morphisms. This proof isolates the reusable algebraic mechanism behind the result. $\square$

Problem 28.19 (Chapter synthesis). Connect the definitions and principal structural theorems of this chapter into one reusable commutative-algebra strategy.

Solution. Derived functors organize Tor, Ext, and similar constructions as systematic corrections to nonexact functors. Resolutions make these corrections computable, while long exact sequences expose their functorial structure. The recurring strategy is to encode the algebraic object by kernels, quotients, localization, finite presentation, or a resolution, apply the corresponding universal or exactness principle, and then recover the desired global or local conclusion. $\square$

28.14 Exercises with complete solutions

Exercise 28.20. State and apply the central principle behind Exact and derived behavior. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Left or right exact functors lose information at kernels or cokernels; derived functors measure that failure in higher degrees. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 28.21. State and apply the central principle behind Left derived functors. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Projective or flat resolutions produce left derived functors of right exact functors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 28.22. State and apply the central principle behind Right derived functors. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Injective resolutions produce right derived functors of left exact functors. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 28.23. State and apply the central principle behind Tor as derived tensor. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Tor_n is the left derived functor of tensor product. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 28.24. State and apply the central principle behind Ext as derived Hom. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Ext^n is the right derived functor of Hom in the second variable or equivalently computed projectively in the first. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 28.25. State and apply the central principle behind Universality. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Derived functors are characterized by effaceability and universal delta-functor properties. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 28.26. State and apply the central principle behind Long exact sequences. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. Connecting morphisms package the failure of exactness into functorial long exact sequences. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Exercise 28.27. State and apply the central principle behind Acyclic resolutions. Give one concrete algebraic consequence.

Hint. Begin from the exact definition or theorem hypothesis in the corresponding section. $\square$

Solution. One may compute derived functors using suitable classes of acyclic objects, not only projectives or injectives. The same principle can then be reused in later localization, support, base-change, resolution, Tor, or Ext arguments. $\square$

Chapter summary

The chapter now contributes a canonical solved layer to the progression from rings and modules through localization, Noetherian methods, integral dependence, and derived functors.